# JJM output definitions and descriptions

June 09, 2022

## Introduction

Organized initially by type of variable.

## Names (strings)

$Like_Comp_names [,1] [1,] “catch_like” [2,] “age_like_fsh” [3,] “length_like_fsh” [4,] “sel_like_fsh” [5,] “ind_like” [6,] “age_like_ind” [7,] “length_like_ind” [8,] “sel_like_ind” [9,] “rec_like” [10,] “fpen” [11,] “post_priors_indq” [12,] “post_priors” [13,] “residual” [14,] “total”

## Scalars

$repl_yld [1] 0

$repl_SSB [1] 0

$FW_N_Chile [1] 0.915323

$FW_SC_Chile_PS [1] 0.910934

$FW_Offshore_Trawl [1] 0.681234

$FW_Chile_AcousCS [1] 0.92536

$FW_Chile_AcousN [1] 0.913439

$FW_DEPM [1] 1.07482

$F40_est [1] 0

$F35_est [1] 0

$rec_like_SRR_stock_1 [1] 2.39911

$rec_like_pen_stock_1 [1] 9.90204

$rec_like_fut_stock_1 [1] -10.21651

## By year and age

$M [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [1,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [2,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [3,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [4,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [5,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [6,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [7,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [8,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23

[9,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [10,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [11,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [12,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [13,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [14,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [15,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [16,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [17,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [18,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [19,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [20,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [21,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [22,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [23,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [24,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [25,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [26,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [27,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [28,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [29,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [30,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [31,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [32,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [33,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [34,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [35,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [36,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [37,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [38,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [39,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [40,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [41,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [42,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [43,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [44,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [45,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [46,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [47,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [48,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [49,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [50,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [51,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 [52,] 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23 0.23

$N [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [1,] 1970 10165.20 6856.33 5023.04 3774.86 2577.03 2058.800 1585.150 1382.070 1203.480 1042.890 897.2050 4685.1600 [2,] 1971 9797.53 8074.92 5444.61 3983.22 2987.59 2028.640 1603.010 1215.080 1065.500 946.631 821.7860 4398.8400 [3,] 1972 10258.90 7781.73 6409.92 4311.84 3145.96 2341.070 1563.930 1207.360 922.956 833.902 742.9230 4097.1900 [4,] 1973 10729.30 8146.44 6174.61 5068.67 3410.08 2478.790 1827.120 1203.260 931.525 724.813 657.6620 3817.1800 [5,] 1974 13096.60 8514.93 6455.22 4858.88 3996.24 2677.630 1922.040 1389.870 918.361 728.145 570.5870 3522.6800 [6,] 1975 18359.80 10370.50 6712.80 4987.75 3789.89 3110.870 2047.390 1428.920 1042.680 714.457 569.3830 3200.7900 [7,] 1976 21782.70 14571.00 8217.08 5281.41 3917.69 2943.020 2353.040 1490.170 1050.880 805.055 556.7950 2938.1900 [8,] 1977 21056.00 17283.60 11539.00 6452.08 4137.79 3025.810 2199.940 1676.730 1075.200 805.444 624.5640 2711.4200 [9,] 1978 21872.00 16563.00 13421.80 8447.74 4891.41 3161.010 2248.230 1557.570 1197.420 816.553 622.7370 2579.2500 [10,] 1979 21926.90 17259.00 12951.30 10071.70 6430.87 3661.560 2224.290 1437.110 1012.970 876.877 615.6670 2414.2400 [11,] 1980 22834.10 17323.70 13533.20 9816.21 7700.57 4801.530 2533.720 1359.930 879.875 713.483 640.5210 2213.2200 [12,] 1981 27808.50 18021.10 13552.50 10172.10 7502.48 5803.450 3403.610 1626.830 873.742 630.080 528.2680 2112.9300 [13,] 1982 32400.70 21973.10 14093.90 10149.90 7706.43 5499.750 3848.580 1926.620 916.463 586.479 452.3200 1896.0500 [14,] 1983 26704.50 25647.00 17330.30 10906.20 7682.17 5389.390 3218.840 1706.050 845.544 541.578 380.4910 1523.5600 [15,] 1984 43440.90 21187.90 20281.60 13498.80 8398.16 5603.330 3456.620 1684.580 859.795 515.743 356.3920 1252.9800 [16,] 1985 52566.60 34449.80 16698.30 15614.70 10117.80 5889.740 3323.540 1544.610 700.700 460.657 300.1890 936.7370 [17,] 1986 29007.90 41705.20 27220.60 12992.90 11816.90 7197.980 3612.360 1575.010 670.251 371.434 270.0710 725.1760 [18,] 1987 27173.30 23029.60 33041.30 21418.90 10056.60 8784.310 4836.110 1944.660 731.111 358.272 219.7990 588.9440 [19,] 1988 32382.10 21559.90 18181.40 25736.80 16261.40 7323.930 5800.230 2521.010 826.026 342.584 186.8000 421.6720 [20,] 1989 28137.60 25658.30 16864.70 13991.60 19221.80 11489.700 4771.380 3146.300 1065.520 334.181 149.5530 265.6250 [21,] 1990 30596.70 22290.30 20080.80 12928.50 10502.30 13589.700 7367.230 2704.720 1520.680 455.523 137.6130 170.9680 [22,] 1991 21202.20 24238.50 17508.60 15369.60 9657.47 7522.500 8904.080 4262.690 1378.080 685.774 184.0700 124.6930 [23,] 1992 21867.00 16797.70 19048.50 13434.30 11458.00 6939.470 4976.320 5065.840 2012.650 572.811 246.2240 110.8600 [24,] 1993 15434.60

17322.90 13186.80 14558.00 9968.50 8118.350 4581.640 2946.030 2381.560 744.381 165.9720 103.4650 [25,] 1994 14473.10 12215.40 13473.60 9791.62 10498.50 6881.770 5293.540 2750.800 1550.630 990.451 220.1310 79.6790 [26,] 1995 18968.30 11443.50 9460.44 9849.29 6740.39 6735.660 4102.240 2933.800 1312.050 532.704 243.4050 73.6786 [27,] 1996 21098.70 14913.80 8569.72 6197.67 5842.43 3612.690 3261.150 1878.570 1219.310 408.987 112.8240 67.1565 [28,] 1997 28551.40 16520.30 10848.60 5326.97 3291.62 2642.470 1564.390 1431.610 784.072 430.049 112.1940 49.3725 [29,] 1998 27067.30 22312.50 11821.60 6502.09 2490.04 1258.150 1050.300 683.688 606.668 281.455 128.0960 48.1248 [30,] 1999 31938.20 21209.30 16255.80 7976.94 3594.53 1265.980 655.764 573.596 368.080 300.066 125.9610 78.8650 [31,] 2000 33038.20 25186.80 16099.20 11519.40 4946.52 2092.180 745.750 401.766 351.904 215.111 164.0500 111.9810 [32,] 2001 21874.50 25952.50 18969.40 11728.20 7459.79 2971.340 1284.600 481.525 262.644 223.507 131.0470 168.1600 [33,] 2002 14907.90 16908.20 18149.70 12869.20 7442.57 4145.860 1689.980 786.888 301.344 159.396 129.0090 172.7050 [34,] 2003 8477.08 11750.10 13059.80 13418.10 8781.92 4546.990 2460.260 1046.420 491.707 180.062 89.4065 169.2340 [35,] 2004 8705.78 6643.14 8877.98 9557.14 9325.24 5558.660 2727.040 1510.890 650.477 291.454 99.8667 143.4480 [36,] 2005 6779.95 6831.10 5085.02 6465.12 6645.59 6021.390 3274.900 1606.680 901.664 370.260 156.7900 130.8930 [37,] 2006 6949.78 5319.53 5203.84 3684.29 4511.39 4424.180 3568.120 1913.310 947.408 510.236 200.3820 155.6920 [38,] 2007 4748.75 5420.97 3886.99 3551.35 2440.14 2960.160 2639.430 1943.070 1044.820 495.952 259.5180 181.1070 [39,] 2008 4247.38 3706.39 3974.79 2473.52 2189.32 1502.530 1726.020 1396.860 939.560 477.875 215.4250 191.3930 [40,] 2009 7545.15 3310.22 2671.49 2530.48 1529.67 1343.190 854.701 915.856 687.688 440.189 211.1940 179.7910 [41,] 2010 9373.33 5897.71 2450.90 1796.17 1496.51 865.611 671.905 396.421 390.507 279.522 170.2960 151.2600 [42,] 2011 5271.78 7300.23 4258.98 1673.81 1085.34 879.201 489.587 374.756 212.267 189.495 135.4100 155.7730 [43,] 2012 8713.50 4135.82 5269.72 2830.12 1187.10 749.142 556.705 316.919 247.440 135.420 123.8580 190.3230 [44,] 2013 7611.43 6890.54 3208.30 3813.39 2092.34 819.737 470.733 356.825 213.814 168.514 93.6608 217.2980 [45,] 2014 10094.60 6025.50 5374.86 2445.37 2854.35 1491.170 541.384 313.293 247.136 150.238 119.4690 220.4560 [46,] 2015 10126.00 7990.37 4694.60 4089.73 1843.81 2077.240 1051.720 370.205 216.126 172.608 105.8190 239.4220 [47,] 2016 15454.20 8020.96 6265.73 3646.84 3067.30 1349.810 1505.590 736.282 249.975 145.874 118.1020 236.2220 [48,] 2017 37164.30 12255.80 6324.31 4900.20 2772.98 2250.560 961.414 1057.380 506.527 170.226 100.3980 243.8650 [49,] 2018 33899.30 29466.40 9670.64 4955.97 3751.74 2064.130 1627.390 674.165 733.474 348.093 117.4730 237.5760 [50,] 2019 20148.00 26881.50 23196.00 7586.38 3835.06 2810.850 1490.970 1135.870 461.745 502.369 240.2620 245.0630 [51,] 2020 14138.10 15976.10 21218.00 18123.00 5915.56 2916.530 2051.510 1032.510 771.936 313.807 350.1000 338.2220 [52,] 2021 15491.30 11215.10 12639.10 16625.70 14189.20 4558.930 2163.530 1443.490 699.803 530.814 222.8060 488.7150

$F_age_1 [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [1,] 1970 8.40087e-05 0.000267102 0.000937201 0.003233490 0.00829580 0.01813470 0.03192130 0.02565710 0.00768231 0.00768231 0.00768231 0.00768231 [2,] 1971 1.24583e-04 0.000396108 0.001389850 0.004795200 0.01230250 0.02689340 0.04733870 0.03804910 0.01139270 0.01139270 0.01139270 0.01139270 [3,] 1972 5.70206e-05 0.000181295 0.000636122 0.002194720 0.00563074 0.01230890 0.02166650 0.01741470 0.00521434 0.00521434 0.00521434 0.00521434 [4,] 1973 7.28237e-05 0.000231540 0.000812421 0.002802980 0.00719129 0.01572020 0.02767130 0.02224110 0.00665948 0.00665948 0.00665948 0.00665948 [5,] 1974 1.41810e-04 0.000450878 0.001582030 0.005458230 0.01400360 0.03061200 0.05388420 0.04331010 0.01296800 0.01296800 0.01296800 0.01296800 [6,] 1975 1.76350e-04 0.000560699 0.001967360 0.006787710 0.01741450 0.03806820 0.06700890 0.05385930 0.01612660 0.01612660 0.01612660 0.01612660 [7,] 1976 2.15779e-04 0.000686059 0.002407220 0.008305300 0.02130800 0.04657940 0.08199070 0.06590100 0.01973220 0.01973220 0.01973220 0.01973220 [8,] 1977 1.88212e-04 0.000598414 0.002099700 0.007244280 0.01858580 0.04062880 0.07151620 0.05748200 0.01721140 0.01721140 0.01721140 0.01721140 [9,] 1978 3.71838e-04 0.001182240 0.004148220 0.014312000 0.03671870 0.08026740 0.14128900 0.11356300 0.03400330 0.03400330 0.03400330 0.03400330 [10,] 1979 3.12516e-04 0.000993631 0.003486420 0.012028700 0.03086070 0.06746180 0.11874800 0.09544560 0.02857850 0.02857850 0.02857850 0.02857850 [11,] 1980 2.15096e-04 0.000683890 0.002399610 0.008279040 0.02124060 0.04643220 0.08173140 0.06569270 0.01966980 0.01966980 0.01966980 0.01966980 [12,] 1981 3.17652e-04 0.001009960 0.003543730 0.012226400 0.03136790 0.06857060 0.12070000 0.09701440 0.02904820 0.02904820 0.02904820 0.02904820 [13,] 1982 6.13468e-04 0.001950500 0.006843840 0.023612300 0.06057950 0.13242700 0.23310300 0.18735900 0.05609950 0.05609950 0.05609950 0.05609950 [14,] 1983 2.37509e-04 0.000755150 0.002649650 0.009141690 0.02345380 0.05127030 0.09024760 0.07253770 0.02171940 0.02171940 0.02171940 0.02171940 [15,] 1984

6.91595e-04 0.002412770 0.009027540 0.028893600 0.05549630 0.10339200 0.16717900 0.14114300 0.06932230 0.06932230 0.06932230 0.06932230 [16,] 1985 6.20728e-04 0.002229390 0.007584920 0.020798500 0.04159240 0.07160030 0.09494920 0.07963110 0.05268460 0.05268460 0.05268460 0.05268460 [17,] 1986 8.61679e-05 0.000346259 0.001094590 0.002369140 0.00370580 0.00586248 0.00851048 0.00776625 0.00564473 0.00564473 0.00564473 0.00564473 [18,] 1987 4.91709e-04 0.002137430 0.008830730 0.019029500 0.02294990 0.02569610 0.03015570 0.03020860 0.02715500 0.02715500 0.02715500 0.02715500 [19,] 1988 5.09400e-04 0.001958170 0.007123370 0.018283700 0.02742440 0.02639010 0.02340720 0.02183920 0.02080000 0.02080000 0.02080000 0.02080000 [20,] 1989 5.77213e-04 0.001975770 0.005668470 0.013643400 0.02784650 0.03190110 0.02332550 0.01823170 0.01659930 0.01659930 0.01659930 0.01659930 [21,] 1990 6.54980e-04 0.002307640 0.005761980 0.012356800 0.02520580 0.03148230 0.02202650 0.01606180 0.01385540 0.01385540 0.01385540 0.01385540 [22,] 1991 9.48846e-04 0.003484410 0.010829100 0.019077000 0.02698530 0.03291990 0.02865450 0.02014660 0.01614700 0.01614700 0.01614700 0.01614700 [23,] 1992 1.16492e-03 0.004945330 0.016035900 0.022598600 0.02822980 0.03135850 0.02573650 0.01883340 0.01552840 0.01552840 0.01552840 0.01552840 [24,] 1993 1.63040e-03 0.008465110 0.036898600 0.035021800 0.02895380 0.02672130 0.02260010 0.01876610 0.01687480 0.01687480 0.01687480 0.01687480 [25,] 1994 1.34512e-03 0.007733030 0.024630500 0.025863400 0.02348710 0.01826440 0.01357320 0.01143520 0.01076100 0.01076100 0.01076100 0.01076100 [26,] 1995 2.03550e-03 0.011577200 0.036618200 0.043538000 0.03184110 0.02223330 0.01650720 0.01411280 0.01361990 0.01361990 0.01361990 0.01361990 [27,] 1996 3.90454e-03 0.019289900 0.067637300 0.095107900 0.05743940 0.03687570 0.02741110 0.02375530 0.02290470 0.02290470 0.02290470 0.02290470 [28,] 1997 1.36078e-03 0.006722330 0.030636400 0.041480100 0.02331220 0.01460770 0.01081560 0.00919953 0.00870801 0.00870801 0.00870801 0.00870801 [29,] 1998 4.02864e-04 0.001914030 0.009218290 0.010938300 0.00729029 0.00458249 0.00320363 0.00262937 0.00242350 0.00242350 0.00242350 0.00242350 [30,] 1999 7.77570e-04 0.003745040 0.015870000 0.016014800 0.01066140 0.00715814 0.00512449 0.00414474 0.00374634 0.00374634 0.00374634 0.00374634 [31,] 2000 1.68665e-03 0.007737650 0.019452500 0.025213000 0.01743710 0.01098140 0.00777983 0.00628100 0.00563539 0.00563539 0.00563539 0.00563539 [32,] 2001 3.91453e-03 0.016210600 0.051986700 0.047856500 0.02841830 0.01935150 0.01446600 0.01188340 0.01070160 0.01070160 0.01070160 0.01070160 [33,] 2002 2.13146e-03 0.007913320 0.021730000 0.020138600 0.01256930 0.00808414 0.00589745 0.00484015 0.00438919 0.00438919 0.00438919 0.00438919 [34,] 2003 3.20439e-03 0.012324900 0.032804300 0.028398900 0.01643750 0.01045690 0.00758696 0.00627918 0.00577129 0.00577129 0.00577129 0.00577129 [35,] 2004 3.65020e-03 0.012563000 0.029502000 0.039704400 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4.69382e-04 4.69382e-04 4.69382e-04 4.69382e-04 4.69382e-04 4.69382e-04 [24,] 1993 0.001054400 0.007864020 0.011834900 0.005775190 0.002294150 1.07162e-03 7.16595e-04 7.16595e-04 7.16595e-04 7.16595e-04 7.16595e-04 7.16595e-04 [25,] 1994 0.001873870 0.010240500 0.026583000 0.011321400 0.003883880 1.88463e-03 1.39717e-03 1.39717e-03 1.39717e-03 1.39717e-03 1.39717e-03 1.39717e-03 [26,] 1995 0.006317870 0.035948800 0.096842700 0.033085300 0.011058500 5.91445e-03 4.99473e-03 4.99473e-03 4.99473e-03 4.99473e-03 4.99473e-03 4.99473e-03 [27,] 1996 0.008674520 0.055401800 0.095009800 0.033542900 0.011670900 6.82111e-03 6.08880e-03 6.08880e-03 6.08880e-03 6.08880e-03 6.08880e-03 6.08880e-03 [28,] 1997 0.012951800 0.080024700 0.126102000 0.042837700 0.016661200 1.00644e-02 8.90969e-03 8.90969e-03 8.90969e-03 8.90969e-03 8.90969e-03 8.90969e-03 [29,] 1998 0.012016300 0.071231700 0.046037600 0.019557500 0.009237810 5.89458e-03 5.08086e-03 5.08086e-03 5.08086e-03 5.08086e-03 5.08086e-03 5.08086e-03 [30,] 1999 0.005695420 0.032234000 0.016162800 0.006457850 0.003272710 2.21406e-03 1.92632e-03 1.92632e-03 1.92632e-03 1.92632e-03 1.92632e-03 1.92632e-03 [31,] 2000 0.008936150 0.039516000 0.021848800 0.007688800 0.003672130 2.50087e-03 2.21305e-03 2.21305e-03 2.21305e-03 2.21305e-03 2.21305e-03 2.21305e-03 [32,] 2001 0.022666400 0.104966000 0.065829200 0.020203700 0.008937870 6.13297e-03 5.62729e-03 5.62729e-03 5.62729e-03 5.62729e-03 5.62729e-03 5.62729e-03 [33,] 2002 0.004912760 0.014349800 0.016178700 0.003553040 0.001451030 1.03935e-03 1.02368e-03 1.02368e-03 1.02368e-03 1.02368e-03 1.02368e-03 1.02368e-03 [34,] 2003 0.009572280 0.032576300 0.022680900 0.006475810 0.002667330 1.81997e-03 1.73049e-03 1.73049e-03 1.73049e-03 1.73049e-03 1.73049e-03 1.73049e-03 [35,] 2004 0.007743960 0.019234200 0.032443700 0.009805880 0.003229050 1.84021e-03 1.55601e-03 1.55601e-03 1.55601e-03 1.55601e-03 1.55601e-03 1.55601e-03 [36,] 2005 0.005363920 0.011800100 0.011051000 0.007804240 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3.64905e-03 3.64905e-03 3.64905e-03 [43,] 2012 0.002914590 0.017573500 0.078947000 0.026262600 0.007142540 2.64162e-03 1.54119e-03 1.54119e-03 1.54119e-03 1.54119e-03 1.54119e-03 1.54119e-03 [44,] 2013 0.001344650 0.007849260 0.024808400 0.019011300 0.003672810 1.13754e-03 6.58785e-04 6.58785e-04 6.58785e-04 6.58785e-04 6.58785e-04 6.58785e-04 [45,] 2014 0.001384040 0.007846030 0.027831500 0.009592090 0.002542870 9.56211e-04 5.80707e-04 5.80707e-04 5.80707e-04 5.80707e-04 5.80707e-04 5.80707e-04 [46,] 2015 0.000716713 0.007089800 0.005484260 0.002031530 0.000720792 3.07584e-04 1.94093e-04 1.94093e-04 1.94093e-04 1.94093e-04 1.94093e-04 1.94093e-04 [47,] 2016 0.000471170 0.003831810 0.003574960 0.001106100 0.000384237 1.77507e-04 1.14918e-04 1.14918e-04 1.14918e-04 1.14918e-04 1.14918e-04 1.14918e-04 [48,] 2017 0.000276993 0.001919210 0.001584450 0.000545480 0.000196914 8.78401e-05 5.57210e-05 5.57210e-05 5.57210e-05 5.57210e-05 5.57210e-05 5.57210e-05 [49,] 2018 0.000831956 0.006647660 0.006469580 0.002521780 0.000860355 3.53311e-04 2.10607e-04 2.10607e-04 2.10607e-04 2.10607e-04 2.10607e-04 2.10607e-04 [50,] 2019 0.000884543 0.004725450 0.012876000 0.004538990 0.001393630 5.21667e-04 2.92084e-04 2.92084e-04 2.92084e-04 2.92084e-04 2.92084e-04 2.92084e-04 [51,] 2020 0.000632373 0.002986620 0.011063000 0.005049830 0.001476070 4.84199e-

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3.21262e-13 [7,] 1976 6.84558e-09 2.17575e-08 7.08967e-08 2.44965e-07 8.84966e-07 2.93080e-06 6.87775e-06 8.11166e-06 4.70985e-06 4.70985e-06 4.70985e-06 4.70985e-06 [8,] 1977 4.70309e-07 1.49480e-06 4.87078e-06 1.68298e-05 6.07994e-05 2.01353e-04 4.72519e-04 5.57291e-04 3.23578e-04 3.23578e-04 3.23578e-04 3.23578e-04 [9,] 1978 9.87003e-06 3.13702e-05 1.02220e-04 3.53194e-04 1.27595e-03 4.22566e-03 9.91642e-03 1.16955e-02 6.79071e-03 6.79071e-03 6.79071e-03 6.79071e-03 [10,] 1979 6.15065e-05 1.95488e-04 6.36996e-04 2.20098e-03 7.95128e-03 2.63327e-02 6.17955e-02 7.28820e-02 4.23172e-02 4.23172e-02 4.23172e-02 4.23172e-02 [11,] 1980 5.58989e-05 1.78922e-04 5.90534e-04 2.14733e-03 8.14436e-03 2.59733e-02 5.59056e-02 6.13713e-02 3.80671e-02 3.80671e-02 3.80671e-02 3.80671e-02 [12,] 1981 7.30855e-05 2.34762e-04 7.71087e-04 2.70868e-03 1.00370e-02 3.21221e-02 6.94468e-02 7.79581e-02 4.94971e-02 4.94971e-02 4.94971e-02 4.94971e-02 [13,] 1982 1.71925e-04 5.58731e-04 1.81933e-03 6.08908e-03 2.12188e-02 6.60897e-02 1.46532e-01 1.74546e-01 1.17774e-01 1.17774e-01 1.17774e-01 1.17774e-01 [14,] 1983 2.10774e-04 7.03136e-04 2.31934e-03 7.51667e-03 2.50071e-02 7.57922e-02 1.61516e-01 1.94398e-01 1.43367e-01 1.43367e-01 1.43367e-01 1.43367e-01 [15,] 1984 2.92546e-04 9.62411e-04 3.20367e-03 1.06494e-02 3.27011e-02 9.14479e-02 1.98807e-01 2.54542e-01 1.95408e-01 1.95408e-01 1.95408e-01 1.95408e-01 [16,] 1985 2.53816e-04 8.23503e-04 2.71394e-03 9.16133e-03 2.79410e-02 7.37739e-02 1.53092e-01 1.94002e-01 1.62782e-01 1.62782e-01 1.62782e-01 1.62782e-01 [17,] 1986 2.64837e-04 8.66258e-04 2.74542e-03 8.85146e-03 2.67372e-02 6.69800e-02 1.36033e-01 1.80093e-01 1.58572e-01 1.58572e-01 1.58572e-01 1.58572e-01 [18,] 1987 3.03449e-04 1.03634e-03 3.19181e-03 9.04845e-03 2.52800e-02 6.54612e-02 1.36660e-01 1.82253e-01 1.68920e-01 1.68920e-01 1.68920e-01 1.68920e-01 [19,] 1988 3.39729e-04 1.15067e-03 3.72893e-03 1.09262e-02 2.74333e-02 6.36129e-02 1.25229e-01 1.77700e-01 1.81084e-01 1.81084e-01 1.81084e-01 1.81084e-01 [20,] 1989 3.14151e-04 1.06320e-03 3.42722e-03 1.01622e-02 2.68032e-02 5.82652e-02 1.02846e-01 1.40593e-01 1.55777e-01 1.55777e-01 1.55777e-01 1.55777e-01 [21,] 1990 2.87841e-04 1.01241e-03 3.28491e-03 9.41240e-03 2.45879e-02 5.48454e-02 9.05478e-02 1.16275e-01 1.29435e-01 1.29435e-01 1.29435e-01 1.29435e-01 [22,] 1991 1.95666e-04 7.50508e-04 2.61414e-03 7.19028e-03 1.63971e-02 3.48423e-02 6.03239e-02 7.50306e-02 8.03374e-02 8.03374e-02 8.03374e-02 8.03374e-02 [23,] 1992 1.67347e-05 6.41887e-05 2.23579e-04 6.14962e-04 1.40240e-03 2.97996e-03 5.15931e-03 6.41714e-03 6.87101e-03 6.87101e-03 6.87101e-03 6.87101e-03 [24,] 1993 9.90062e-16 3.79754e-15 1.32274e-14 3.63825e-14 8.29687e-14 1.76300e-13 3.05236e-13 3.79651e-13 4.06503e-13 4.06503e-13 4.06503e-13 4.06503e-13 [25,] 1994 1.00514e-15 3.85537e-15 1.34288e-14 3.69365e-14 8.42322e-14 1.78985e-13 3.09884e-13 3.85433e-13 4.12694e-13 4.12694e-13 4.12694e-13 4.12694e-13 [26,] 1995 1.29914e-15 4.98306e-15 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$SSB_fut_5 [,1] [,2] [,3] [,4] [,5] [1,] 2022 12054.7 1195.80 9890.16 14692.9 [2,] 2023 13310.0 1337.52 10892.20

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$Catch_fut_1 [,1] [,2] [1,] 2022 981.166 [2,] 2023 1213.500 [3,] 2024 1428.560 [4,] 2025 1498.900 [5,] 2026 1439.600 [6,] 2027 1394.910 [7,] 2028 1421.510 [8,] 2029 1395.630 [9,] 2030 1317.220 [10,] 2031 1245.240 [11,] 2032 1193.440 [12,] 2033 1144.060 [13,] 2034 1108.910 [14,] 2035 1083.270 [15,] 2036 1063.870 [16,] 2037 1048.440 [17,] 2038 1035.700 [18,] 2039 1025.050 [19,] 2040 1016.120 [20,] 2041 1008.420

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$Catch_fut_5 [,1] [,2] [1,] 2022 0 [2,] 2023 0 [3,] 2024 0 [4,] 2025 0 [5,] 2026 0 [6,] 2027 0 [7,] 2028 0 [8,] 2029 0 [9,] 2030 0 [10,] 2031 0 [11,] 2032 0 [12,] 2033 0 [13,] 2034 0 [14,] 2035 0 [15,] 2036 0 [16,] 2037 0 [17,] 2038 0 [18,] 2039 0 [19,] 2040 0 [20,] 2041 0

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0.01541560 0.04998710 0.1104280 0.1823650 0.2755500 0.1339980 0.0829113 0.05064850 0.03809710 0.02722350 0.03131730 [38,] 2012 0.00267917 0.00736416 0.05058750 0.1452310 0.2221390 0.2411540 0.1611540 0.0647713 0.04087820 0.01928960 0.01764270 0.02711010 [39,] 2013 0.00170407 0.00984269 0.02899630 0.1845170 0.3092040 0.2108220 0.1115460 0.0594132 0.02795400 0.01968190 0.01093930 0.02537970 [40,] 2014 0.00183096 0.00778761 0.04940020 0.1223060 0.3297280 0.2373170 0.1049600 0.0523056 0.03378790 0.01856700 0.01476440 0.02724470 [41,] 2015 0.00165719 0.01014220 0.05030870 0.2342250 0.1691280 0.2123650 0.1485690 0.0647900 0.03505800 0.02458440 0.01507170 0.03410080 [42,] 2016 0.00213195 0.00821013 0.04716900 0.1447090 0.2487770 0.1528420 0.1910910 0.1052470 0.03530190 0.01881660 0.01523420 0.03047070 [43,] 2017 0.00464519 0.01049030 0.03884210 0.1452290 0.1759470 0.2197920 0.1257890 0.1448100 0.06793470 0.02200920 0.01298090 0.03153030 [44,] 2018 0.00381727 0.01876510 0.03580500 0.1009570 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2000 0.00000e+00 0.00000e+00 0.00000e+00 1.09923e-04 1.92366e-04 1.92366e-04 1.92366e-04 4.94655e-04 4.89159e-03 4.10838e-02 7.60120e-02 6.79601e-02 5.10044e-02 3.68518e-02 3.67968e-02 9.59631e-02 [22,] 2001 0.00000e+00 0.00000e+00 1.66978e-04 3.44392e-04 3.44392e-04 1.66978e-04 4.27881e-04 1.96199e-03 4.79018e-03 9.83083e-03 1.56542e-02 1.40262e-02 1.36087e-02 2.13001e-02 4.21411e-02 6.51423e-02 [23,] 2002 4.81493e-04 9.02799e-04 1.98616e-03 4.27325e-03 4.87511e-03 6.07884e-03 4.27325e-03 2.82877e-03 2.10653e-03 2.70840e-03 2.40746e-03 3.43063e-03 5.59735e-03 7.40295e-03 8.42612e-03 4.09269e-03 [24,] 2003 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 1.01600e-04 1.01600e-04 4.06401e-04 1.57480e-03 3.09881e-03 8.07722e-03 1.41224e-02 2.30124e-02 4.59741e-02 7.20853e-02 7.89942e-02 5.95885e-02 [25,] 2004 5.83686e-04 1.09441e-03 3.42916e-03 3.72100e-03 3.28323e-03 2.91843e-03 4.01284e-03 3.50212e-03 1.38625e-03 3.64804e-04 7.29607e-05 0.00000e+00 7.29607e-05 7.29607e-05 7.29607e-05 7.29607e-05 [26,] 2005 4.81876e-04 3.80147e-03 1.17792e-02 1.21004e-02 8.08481e-03 4.97939e-03 3.48022e-03 5.08647e-03 3.11078e-02 6.14124e-02 7.54404e-02 8.31504e-02 9.65358e-02 9.88381e-02 6.05022e-02 2.52182e-02 [27,] 2006 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 5.00000e-04 4.90000e-03 1.50000e-02 3.04000e-02 6.32000e-02 8.01000e-02 7.49000e-02 7.75000e-02 [28,] 2007 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 1.97746e-04 3.02435e-04 3.02435e-04 9.01488e-04 1.50054e-03 1.19811e-03 1.09923e-03 1.59941e-03 2.50090e-03 7.70045e-03 4.50046e-02 8.83109e-02 [29,] 2008 0.00000e+00 0.00000e+00 9.17221e-05 1.83444e-04 5.96194e-04 1.00894e-03 5.04471e-04 1.83444e-04 9.17221e-05 1.83444e-04 5.04471e-04 1.00894e-03 9.17221e-04 1.42169e-03 3.39372e-03 5.50332e-03 [30,] 2009 8.36468e-03 7.96872e-03 1.12354e-02 1.12354e-02 2.67274e-03 1.03940e-03 0.00000e+00 0.00000e+00 0.00000e+00 1.33637e-03 9.00812e-03 3.60820e-02 4.06850e-02 3.74183e-02 2.31142e-02 1.55415e-02 [31,] 2010 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 1.86199e-02 0.00000e+00 1.86199e-02 2.53012e-01 4.93976e-01 1.77437e-01 [32,] 2011 0.00000e+00 0.00000e+00 0.00000e+00 8.15066e-05 8.15066e-05 8.15066e-05 8.15066e-05 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 8.15066e-05 0.00000e+00 1.63013e-04 6.73122e-03 [33,] 2012 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 3.84144e-04 3.68648e-03 8.07589e-03 [34,] 2013 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 1.24399e-03 7.13057e-03 1.13031e-02 6.32162e-03 2.09200e-03 9.44914e-03 1.71929e-02 [35,] 2014 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 3.17692e-05 1.16487e-04 1.90615e-04 1.90615e-04 5.29487e-05 2.11795e-05 1.05897e-05 2.11795e-05 5.29487e-05 1.27077e-04 1.05897e-04 1.05897e-04 [36,] 2015 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 2.52325e-04 1.05500e-04 2.27409e-03 3.68973e-03 4.78372e-03 8.14524e-03 1.43651e-02 3.39367e-02 1.03460e-01 1.28175e-01 1.23798e-01 [37,] 2016 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 8.92593e-05 9.41931e-04 1.43326e-03 3.89820e-03 9.87281e-03 2.46695e-02 5.30516e-02 6.47254e-02 6.13004e-02 [38,] 2017 0.00000e+00 0.00000e+00 0.00000e+00 7.99826e-05 9.14451e-04 4.97200e-03 1.15174e-02 2.09267e-02 2.83460e-02 3.64257e-02 4.85987e-02 6.38203e-02 7.86123e-02 7.56519e-02 7.14347e-02 6.72244e-02 [39,] 2018 0.00000e+00 0.00000e+00 1.02994e-05 5.59033e-05 2.15974e-04 4.10719e-04 1.35555e-03 3.37312e-03 5.62573e-03 6.37533e-03 6.57317e-03 9.25798e-03 1.94857e-02 3.47183e-02 7.08800e-02 1.10017e-01 [40,] 2019 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 9.61690e-06 1.96518e-04 2.57239e-04 2.80319e-04 3.62640e-04 8.79803e-04 2.68857e-03 4.91420e-03 7.40368e-03 1.05160e-02 1.47424e-02 [41,] 2020 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 0.00000e+00 1.08312e-05 9.94636e-05 6.31589e-05 1.73537e-04 6.05095e-04 1.20633e-03 2.34271e-03 4.09504e-03 6.12878e-03 [42,] 2021 3.63176e-03 6.43878e-03 5.60095e-03 7.77433e-03 5.34281e-03 2.31051e-03 1.01389e-03 0.00000e+00 0.00000e+00 6.12915e-06 4.07796e-05 2.99760e-04 1.01101e-03 2.64386e-03 5.38869e-03 5.90486e-03 [,18] [,19] [,20] [,21] [,22] [,23] [,24] [,25] [,26] [,27] [,28] [,29] [,30] [,31] [,32] [,33] [,34] [1,] 0.002348640 0.00234864 0.01304800 0.02635700 0.03496870 0.05454070 0.08298540 0.122912000 1.09342e-01 0.089509400 6.55010e-02 0.048016700 3.49687e-02 0.021920700 0.013048000 0.008611690 0.008611690 [2,] 0.005323870 0.01242240 0.02484470 0.08961850 0.11446300 0.09316770 0.09671690 0.099378900 9.31677e-02 0.077196100 6.92103e-02 0.055013300 4.43656e-02 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0.04072600 0.14072200 [3,] 1981 0.00256291 0.00530873 0.0128446 0.0340524 0.0916187 0.2163760 0.2551110 0.1365830 0.0504816 0.0375708 0.03149990 0.12599100 [4,] 1982 0.00333177 0.00733031 0.0151707 0.0361810 0.0922107 0.1887500 0.2588450 0.1536850 0.0562969 0.0376081 0.02900510 0.12158500 [5,] 1983 0.00308590 0.00987111 0.0218425 0.0443047 0.1011790 0.2025550 0.2350340 0.1474380 0.0587365 0.0389644 0.02737480 0.10961400 [6,] 1984 0.00554659 0.00887323 0.0279588 0.0610707 0.1130560 0.1951820 0.2307610 0.1396120 0.0611179 0.0380591 0.02629980 0.09246340 [7,] 1985 0.00723796 0.01535990 0.0243560 0.0758682 0.1456010 0.2088530 0.2178300 0.1234670 0.0513164 0.0353068 0.02300780 0.07179560 [8,] 1986 0.00405893 0.01906860 0.0393154 0.0600271 0.1617630 0.2353560 0.2167440 0.1171660 0.0467298 0.0271154 0.01971570 0.05293930 [9,] 1987 0.00380414 0.01098450 0.0482259 0.0875452 0.1125900 0.2432100 0.2509790 0.1230520 0.0447418 0.0229843 0.01410080 0.03778270 [10,] 1988 0.00456433 0.01022950 0.0277378 0.1134180 0.1752640 0.1762200 0.2526840 0.1396670 0.0457706 0.0196121 0.01069380 0.02413960 [11,] 1989 0.00382727 0.01174280 0.0246349 0.0599960 0.2113160 0.2623330 0.1817410 0.1525720 0.0542747 0.0167512 0.00749655 0.01331480 [12,] 1990 0.00400431 0.01021870 0.0295014 0.0537957 0.1119080 0.3097940 0.2618660 0.1165970 0.0691641 0.0197633 0.00597049 0.00741760 [13,] 1991 0.00292665 0.01278340 0.0317972 0.0757269 0.1066340 0.1697800 0.3246450 0.1778960 0.0582743 0.0272618 0.00731741 0.00495697 [14,] 2000 0.01421720 0.04606160 0.1213910 0.2358440 0.1898240 0.1356530 0.0754827 0.0542069 0.0537409 0.0322260 0.02457650 0.01677600 [15,] 2001 0.00718547 0.03811000 0.1264330 0.2369450 0.2513700 0.1367850 0.0803696 0.0414044 0.0276376 0.0229870 0.01347780 0.01729480 [16,] 2002 0.00434936 0.02398640 0.1259620 0.2780360 0.2259100 0.1495390 0.0791503 0.0507519 0.0250585 0.0128789 0.01042380 0.01395430 [17,] 2003 0.00239387 0.01551850 0.0814356 0.2637780 0.2477850 0.1564760 0.1062000 0.0595285 0.0358497 0.0127379 0.00632479 0.01197200 [18,] 2006 0.00271191 0.00837219 0.0295396 0.0567676 0.1410320 0.2365740 0.2355960 0.1384310 0.0793462 0.0421883 0.01656830 0.01287320 [19,] 2007 0.00269074 0.01107120 0.0245186 0.0561470 0.0800642 0.1797310 0.2298020 0.1999420 0.1151210 0.0534367 0.02796190 0.01951360 [20,] 2008 0.00353276 0.01033810 0.0320350 0.0480840 0.0826410 0.1027270 0.2089140 0.2256250 0.1491680 0.0739669 0.03334410 0.02962440 [21,] 2009 0.00978935 0.01452700 0.0344890 0.0686014 0.0746242 0.1065060 0.1197250 0.2084000 0.1663950 0.1043010 0.05004150 0.04260070 [22,] 2010 0.01742330 0.03722050 0.0551110 0.0765348 0.0980593 0.0942862 0.1210500 0.1127260 0.1527100 0.1092270 0.06654540 0.05910690 [23,] 2011 0.01193120 0.04829530 0.0794039 0.0765460 0.0889667 0.1207510 0.1044140 0.1235690 0.1052020 0.0949772 0.06786910 0.07807500 [24,] 2012 0.02139700 0.02894020 0.0937890 0.1169190 0.0910719 0.0931091 0.1061240 0.0961222 0.1245590 0.0686642 0.06280190 0.09650280 [25,] 2013 0.01681400 0.04370550 0.0578334 0.1676540 0.1695480 0.1121980 0.0933268 0.0956448 0.0748098 0.0592085 0.03290830 0.07634890 [26,] 2014 0.02229670 0.04024450 0.1163680 0.1119400 0.1888290 0.1557470 0.0883917 0.0707395 0.0686818 0.0419186 0.03333360 0.06151040 [27,] 2015 0.02524880 0.05466210 0.0904637 0.1640670 0.1120260 0.1798510 0.1359820 0.0689742 0.0494594 0.0397535 0.02437120 0.05514160 [28,] 2016 0.04960640 0.06325500 0.1158110 0.1284750 0.1561850 0.0908715 0.1328760 0.1007430 0.0538622 0.0315885 0.02557460 0.05115300 [29,] 2017 0.15664400 0.11241700 0.1020360 0.1207400 0.0873351 0.0870778 0.0539599 0.1008770 0.0886706 0.0298577 0.01760990 0.04277400 [30,] 2018 0.11624700 0.18833600 0.1301170 0.1045990 0.0902875 0.0556877 0.0627884 0.0497423 0.1030500 0.0490827 0.01656420 0.03349930 [31,] 2019 0.09263860 0.20725400 0.2058770 0.0882221 0.0591773 0.0548391 0.0453685 0.0678732 0.0564881 0.0621860 0.02974090 0.03033530 [32,] 2020 0.08676970 0.17780700 0.1958520 0.1459240 0.0601655 0.0430822 0.0518415 0.0537337 0.0797382 0.0329068 0.03671260 0.03546700

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$pobs_ind_1 [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [1,] 1997 0.003627810 0.012991500 0.3851850 0.41631500 0.118737000 0.0327483 0.00642220 0.00563781 0.00627513 0.003676830 0.003725860 0.004657320 [2,] 1998 0.000134511 0.005335610 0.4752280 0.44168900 0.040891400 0.0105815 0.00636686 0.01080570 0.00744294 0.001076090 0.000179348 0.000269022 [3,] 1999 0.000000000 0.019941000 0.2932690 0.43110800 0.107244000 0.0553977 0.02059660 0.01059880 0.02731640 0.019503900 0.011582200 0.003441870 [4,] 2000 0.000000000 0.000000000 0.1001430 0.44729500 0.311396000 0.1103590 0.02245190 0.00447454 0.00166310 0.000791954 0.000633563 0.000791954 [5,] 2001 0.000000000 0.008989920 0.2117770 0.46086200

0.228729000 0.0575355 0.01374170 0.00773775 0.00568291 0.003307010 0.001412700 0.000224748 [6,] 2002 0.000000000 0.000127113 0.0633024 0.28181000 0.232999000 0.3003690 0.07181900 0.03559170 0.01016910 0.002669380 0.000889793 0.000254227 [7,] 2003 0.000000000 0.000110047 0.0303731 0.23462100 0.339936000 0.2578410 0.07483220 0.02200950 0.01716740 0.012655400 0.005942560 0.004511940 [8,] 2004 0.000000000 0.000000000 0.0121041 0.06677430 0.184890000 0.2296750 0.17238200 0.11377800 0.09178940 0.090074600 0.032277600 0.006253780 [9,] 2005 0.000000000 0.000000000 0.0000000 0.09407560 0.207588000 0.4081790 0.18480800 0.05450160 0.02487950 0.004975900 0.006375370 0.014616700 [10,] 2006 0.000000000 0.000000000 0.0000000 0.00121175 0.181662000 0.4307790 0.26507100 0.05160050 0.03251540 0.018984100 0.011107700 0.007068570 [11,] 2007 0.000000000 0.000000000 0.0000000 0.00000000 0.000192086 0.0651172 0.21379200 0.20322700 0.18747600 0.166923000 0.084325800 0.078947400 [12,] 2008 0.000000000 0.000000000 0.0000000 0.00000000 0.000000000 0.1004180 0.33159000 0.26778200 0.12238500 0.091004200 0.035564900 0.051255200 [13,] 2009 0.000000000 0.000000000 0.0000000 0.00000000 0.000000000 0.0000000 0.00865266 0.32756500 0.20890000 0.176761000 0.156984000 0.121137000

$phat_ind_1 [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [1,] 1997 0.05148400 0.0910591 0.1674150 0.1734840 0.154302 0.1464340 0.0866629 0.0713824 0.03427110 0.01708650 0.00445764 0.00196164 [2,] 1998 0.05119660 0.1298290 0.2008240 0.2379360 0.137740 0.0817859 0.0668952 0.0394648 0.03175390 0.01388300 0.00631844 0.00237379 [3,] 1999 0.05057580 0.1048220 0.2353340 0.2556830 0.175650 0.0723098 0.0365858 0.0291927 0.01724470 0.01343350 0.00563910 0.00353068 [4,] 2000 0.04151620 0.0986191 0.1873950 0.2988270 0.194665 0.0966380 0.0338014 0.0166882 0.01356670 0.00800754 0.00610679 0.00416851 [5,] 2001 0.02344510 0.0846028 0.1840570 0.2590500 0.244050 0.1141520 0.0489110 0.0168651 0.00853133 0.00698544 0.00409573 0.00525566 [6,] 2002 0.01490700 0.0531634 0.1689080 0.2711300 0.234388 0.1500230 0.0598137 0.0254884 0.00899975 0.00455542 0.00368700 0.00493577 [7,] 2003 0.00809520 0.0350444 0.1158560 0.2726920 0.268695 0.1582100 0.0830733 0.0323773 0.01401090 0.00490324 0.00243461 0.00460839 [8,] 2004 0.00875719 0.0209724 0.0827475 0.2045270 0.302887 0.2021150 0.0952817 0.0483872 0.01918700 0.00825001 0.00282687 0.00406050 [9,] 2005 0.00940180 0.0231659 0.0404775 0.1125450 0.217077 0.2818440 0.1714170 0.0790831 0.03896920 0.01464220 0.00620039 0.00517628 [10,] 2006 0.01157940 0.0213542 0.0486637 0.0755868 0.176424 0.2501260 0.2184550 0.1098580 0.04774320 0.02368270 0.00930075 0.00722646 [11,] 2007 0.01054320 0.0290321 0.0470338 0.0942054 0.123803 0.2208330 0.2127310 0.1412850 0.06630610 0.02871500 0.01502580 0.01048590 [12,] 2008 0.01312210 0.0274413 0.0669791 0.0914783 0.154425 0.1544720 0.1939060 0.1424830 0.08385700 0.03880290 0.01749230 0.01554090 [13,] 2009 0.03233810 0.0343403 0.0638100 0.1272780 0.144548 0.1813580 0.1257890 0.1219140 0.08015570 0.04685300 0.02247910 0.01913660

$sdnr_age_ind_1 [,1] [,2] [1,] 1997 0.625800 [2,] 1998 0.613533 [3,] 1999 0.358156 [4,] 2000 0.372305 [5,] 2001 0.352340 [6,] 2002 0.338986 [7,] 2003 0.288389 [8,] 2004 0.747816 [9,] 2005 0.252290 [10,] 2006 0.355137 [11,] 2007 0.842519 [12,] 2008 0.541003 [13,] 2009 1.020390

$pobs_ind_2 [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [1,] 2006 0.120517000 0.41630300 0.2812350 0.1595430 0.02240190 0.00000000 0.00000e+00 0.000000000 0.000000000 0.0000e+00 0.000000000 0 [2,] 2007 0.000197653 0.04998740 0.1747270 0.5471230 0.21151100 0.01429620 2.15753e-03 0.000000000 0.000000000 0.0000e+00 0.000000000 0 [3,] 2008 0.004403320 0.48867500 0.4764580 0.0122276 0.01344510 0.00474859 4.25711e-05 0.000000000 0.000000000 0.0000e+00 0.000000000 0 [4,] 2009 0.000000000 0.41167900 0.4207720 0.1581690 0.00296052 0.00544525 9.74254e-04 0.000000000 0.000000000 0.0000e+00 0.000000000 0 [5,] 2010 0.000000000 0.00437424 0.0742794 0.7085950 0.17026600 0.03359700 8.88834e-03 0.000000000 0.000000000 0.0000e+00 0.000000000 0 [6,] 2011 0.000000000 0.26986500 0.1238550 0.5046410 0.03661240 0.05090040 1.33138e-02 0.000812203 0.000000000 0.0000e+00 0.000000000 0 [7,] 2012 0.000576041 0.43025200 0.0506772 0.1715050 0.24570200 0.09314050 7.52528e-03 0.000622380 0.000000000 0.0000e+00 0.000000000 0 [8,] 2013 0.106694000 0.34501800 0.0216584 0.1725580 0.16461800 0.14653500 3.76268e-02 0.005291640 0.000000000 0.0000e+00 0.000000000 0 [9,] 2014 0.298696000 0.65585200 0.0090278 0.0132180 0.00328290 0.01230060 7.62247e-03 0.000000000 0.000000000 0.0000e+00 0.000000000 0 [10,] 2015 0.624619000 0.09750090 0.0525165 0.1525520 0.04136900 0.02389620 6.92884e-03 0.000397025 0.000220799 0.0000e+00 0.000000000 0 [11,] 2016 0.696717000 0.04076110 0.1040030 0.0993868 0.03641860 0.01174040 4.76302e-03 0.004378920 0.001167790 0.0000e+00 0.000663099 0 [12,] 2017 0.194674000 0.56910700 0.0707911 0.0990881 0.05412390 0.01031460 1.37248e-03 0.000000000 0.000528294 0.0000e+00 0.000000000 0 [13,] 2018 0.781022000 0.15395900 0.0175638 0.0328232 0.01071590 0.00210101 1.02957e-03 0.000392988 0.000307547 8.5043e-05 0.000000000 0 [14,] 2019 0.037595300 0.12149300 0.1968960 0.3930650 0.19935800 0.03665180 1.15804e-02 0.002592720 0.000767366

0.0000e+00 0.000000000 0 [15,] 2020 0.190767000 0.03975320 0.0134710 0.0656644 0.25356800 0.31179600 1.07987e-01 0.016993800 0.000000000 0.0000e+00 0.000000000 0 [16,] 2021 0.321960000 0.29501900 0.0128543 0.0048951 0.13397300 0.14024800 4.59276e-02 0.029159200 0.015317300 0.0000e+00 0.000645437 0

$phat_ind_2 [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [1,] 2006 0.113098 0.137334 0.1740480 0.1319220 0.1430780 0.1169940 0.0865243 0.04944040 0.02503170 0.01326980 0.005211380 0.00404911 [2,] 2007 0.101177 0.183664 0.1604900 0.1567470 0.0960454 0.1004970 0.0817838 0.05936990 0.03227990 0.01479840 0.007743570 0.00540393 [3,] 2008 0.116322 0.159321 0.2112880 0.1408730 0.1105660 0.0643109 0.0689798 0.05575750 0.03811430 0.01861730 0.008392650 0.00745639 [4,] 2009 0.235199 0.165219 0.1687440 0.1576530 0.0820010 0.0586462 0.0346697 0.03682190 0.02813880 0.01742740 0.008361320 0.00711805 [5,] 2010 0.271415 0.268615 0.1461300 0.1061500 0.0771176 0.0390082 0.0296125 0.01795370 0.01725620 0.01243610 0.007576560 0.00672964 [6,] 2011 0.147874 0.320037 0.2389880 0.1088430 0.0615919 0.0419120 0.0235209 0.01945970 0.01134450 0.01041920 0.007445390 0.00856500 [7,] 2012 0.319415 0.176377 0.2289110 0.1349990 0.0540388 0.0298015 0.0206393 0.01139810 0.00864257 0.00475268 0.004346910 0.00667955 [8,] 2013 0.255847 0.270456 0.1333230 0.1683730 0.0895852 0.0311407 0.0164979 0.01201700 0.00702242 0.00553105 0.003074180 0.00713226 [9,] 2014 0.296208 0.206275 0.1947210 0.0948389 0.1085720 0.0522399 0.0169432 0.00918096 0.00705080 0.00428176 0.003404850 0.00628295 [10,] 2015 0.267166 0.247133 0.1554940 0.1419020 0.0633360 0.0668975 0.0301628 0.00957645 0.00538528 0.00431511 0.002645420 0.00598545 [11,] 2016 0.400655 0.191081 0.1514240 0.0948512 0.0791319 0.0307515 0.0288297 0.01202060 0.00375203 0.00218823 0.001771630 0.00354352 [12,] 2017 0.562609 0.170624 0.0894126 0.0748630 0.0422856 0.0303242 0.0107380 0.01015180 0.00447416 0.00149468 0.000881551 0.00214126 [13,] 2018 0.407869 0.325362 0.1087490 0.0607063 0.0457111 0.0220823 0.0143887 0.00508955 0.00513428 0.00242928 0.000819826 0.00165801 [14,] 2019 0.241729 0.296656 0.2592410 0.0932595 0.0471798 0.0302472 0.0130599 0.00849592 0.00320236 0.00352433 0.001685540 0.00171922 [15,] 2020 0.178342 0.185660 0.2498410 0.2349530 0.0773415 0.0334364 0.0191394 0.00809880 0.00568226 0.00235009 0.002621880 0.00253293 [16,] 2021 0.200420 0.133716 0.1531090 0.2207270 0.1908250 0.0538942 0.0208310 0.01171800 0.00527949 0.00405043 0.001700150 0.00372920

$sdnr_age_ind_2 [,1] [,2] [1,] 2006 0.724413 [2,] 2007 0.900430 [3,] 2008 0.898833 [4,] 2009 0.802775 [5,] 2010 1.464200 [6,] 2011 0.932664 [7,] 2012 0.943902 [8,] 2013 0.603038 [9,] 2014 0.867978 [10,] 2015 0.638932 [11,] 2016 0.506995 [12,] 2017 0.879281 [13,] 2018 0.614434 [14,] 2019 0.937532 [15,] 2020 1.287040 [16,] 2021 0.660742

$pobs_ind_4 [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [1,] 2001 0 0.1505230 0.355499000 0.3659920 0.0570770 0.0278230 0.0166938 0.0159386 0.00723399 0.00230534 0.000556461 0.000357725 [2,] 2003 0 0.0156024 0.152846000 0.2418380 0.0970818 0.1626700 0.1149960 0.1155740 0.06327650 0.02427040 0.010112700 0.001733600 [3,] 2004 0 0.0152921 0.152544000 0.3510900 0.1893960 0.0901228 0.0507646 0.0732013 0.04712960 0.02105790 0.008272750 0.001128100 [4,] 2005 0 0.0000000 0.000000000 0.0070898 0.3764350 0.2424040 0.1637410 0.1134370 0.04591490 0.03173530 0.019243800 0.000000000 [5,] 2006 0 0.0000000 0.000973394 0.0350422 0.1998700 0.3069440 0.2389680 0.1377350 0.04931860 0.01849450 0.012654100 0.000000000 [6,] 2008 0 0.0000000 0.000000000 0.0408277 0.1171700 0.2175620 0.2746090 0.2002240 0.09032440 0.05425060 0.002516780 0.002516780

$phat_ind_4 [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [1,] 2001 0.0460903 0.1380900 0.2345720 0.2380880 0.171742 0.0802461 0.041729 0.0179064 0.0104432 0.00901947 0.00528833 0.00678600 [2,] 2003 0.0124006 0.0403574 0.1036690 0.2146680 0.227931 0.1685720 0.120832 0.0604761 0.0279996 0.00947860 0.00470642 0.00890861 [3,] 2004 0.0129351 0.0233910 0.0712308 0.1552090 0.249489 0.2061710 0.131509 0.0857850 0.0363980 0.01519590 0.00520688 0.00747914 [4,] 2005 0.0109680 0.0260907 0.0442459 0.1146420 0.198333 0.2442960 0.170821 0.0983915 0.0546818 0.02112000 0.00894346 0.00746628 [5,] 2006 0.0133599 0.0234685 0.0515548 0.0748367 0.158902 0.2154350 0.210159 0.1316290 0.0644911 0.03307960 0.01299120 0.01009380 [6,] 2008 0.0148141 0.0293682 0.0676676 0.0864002 0.132753 0.1279220 0.180821 0.1607290 0.1063540 0.05032720 0.02268740 0.02015650

$sdnr_age_ind_4 [,1] [,2] [1,] 2001 0.889498 [2,] 2003 0.788795 [3,] 2004 1.124670 [4,] 2005 0.955554 [5,] 2006 0.629107 [6,] 2008 0.820355

$Obs_catch_1 [1] 101.690 143.450 64.460 83.200 164.760 207.330 257.700 226.230 398.410 344.050 288.810 474.820 789.910 301.930 727.000 511.150 55.210 313.310 325.460 338.600 323.090 346.250 304.240 379.470

222.250 230.180 [27] 278.440 104.200 30.270 55.650 118.730 248.100 108.730 143.280 158.660 165.630 155.260 172.701 167.258 134.022 169.012 30.825 13.256 16.361 18.219 34.886 24.657 35.002 11.551 11.875 44.155 61.359

$Pred_catch_1 [1] 101.7650 143.6110 64.4934 83.2541 164.9610 207.6580 258.2650 226.5880 399.3560 344.7620 289.0990 475.4310 791.8060 302.0620 728.1350 510.8740 55.2012 312.5880 324.3790 337.4790 322.4780 346.0140 304.2480 [24] 379.5400 222.2970 230.1720 277.8390 104.0500 30.2570 55.6059 118.6700 248.2130 108.7320 143.0870 158.3590 165.4370 155.5440 174.9850 169.5060 134.6780 168.5930 30.8442 13.2673 16.3765 18.2344 34.9130 [47] 24.6682 35.0149 11.5517 11.8748 44.1514 61.3576

$Obs_catch_2 [1] 10.310 14.990 22.550 38.390 28.750 53.880 84.570 114.570 188.270 253.460 273.450 586.090 704.770 563.340 699.300 945.840 1129.110 1456.730 1812.790 2051.520 2148.790 2674.270 2907.820 [24] 2856.780 3819.190 4174.020 3604.890 2812.870 1582.640 1164.040 1115.570 1401.840 1410.270 1278.020 1292.940 1264.810 1224.690 1130.080 728.850 700.905 295.796 216.470 214.204 214.999 254.295 250.327 [47] 295.160 311.863 415.149 432.447 517.665 567.267

$Pred_catch_2 [1] 10.3109 14.9921 22.5549 38.4033 28.7566 53.9044 84.6327 114.6810 188.4550 253.7650 273.6730 586.8430 706.1560 563.8180 700.5980 945.9210 1127.2600 1448.5000 1789.5900 2011.8000 2102.5900 [22] 2610.1400 2864.5400 2844.4900 3800.9300 4149.2000 3564.6000 2752.4700 1556.5800 1147.8500 1108.5900 1401.0700 1413.2100 1277.5000 1284.0000 1259.8000 1232.3600 1150.0800 737.6750 709.6530 297.2050 217.5730 [43] 215.9230 217.6740 257.3600 252.1980 297.4540 313.8550 416.6880 432.4710 517.3970 567.1470

$Obs_catch_3 [1] 4.710 9.190 18.780 42.780 129.210 37.900 54.150 504.990 386.790 333.810 414.300 445.640 143.720 110.690 200.670 114.620 51.030 46.300 244.230 316.250 370.820 213.450 111.680 133.350 233.350 550.990 [27] 495.520 680.050 412.850 203.750 303.700 857.740 154.820 217.730 187.370 80.660 277.570 255.360 169.537 76.629 22.172 326.394 187.396 80.586 74.532 22.447 15.088 8.867 57.163 135.784 140.116 129.324

$Pred_catch_3 [1] 4.71005 9.19018 18.78070 42.78340 129.23900 37.90200 54.15130 504.96200 386.76900 333.77800 414.28400 445.61900 143.70500 110.67100 200.68100 114.60400 51.02210 46.29210 244.09200 316.09900 370.82400 [22] 213.47400 111.67700 133.35800 233.42700 550.72600 494.11900 676.97900 411.61900 203.55200 303.52600 857.99600 154.74100 217.46000 187.21500 80.64790 278.34800 257.75900 170.21700 76.67070 22.16370 327.75400 [43] 188.31900 80.69740 74.55700 22.44740 15.08700 8.86651 57.15630 135.77000 140.08500 129.32100

$Obs_catch_4 [1] 0.0000 0.0000 5.5000 0.0000 0.0000 0.0000 0.0400 2.2700 51.2900 370.2900 339.8000 438.1200 733.2000 894.3000 1059.9300 799.3200 837.5000 863.4200 863.2200 875.8200 872.0600 [22] 543.6600 37.9300 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0100 2.3200 20.0900 76.2600 158.2000 295.4400 243.5800 362.6300 438.8310 406.9860 371.9180 239.5930 60.8913 [43] 39.9179 41.1770 63.6520 86.6720 54.1630 49.1130 42.4600 51.4390 4.7390 55.3090

$Pred_catch_4 [1] 2.25082e-09 2.24658e-09 5.50025e+00 2.25013e-09 2.24429e-09 2.24651e-09 4.00000e-02 2.27006e+00 5.13123e+01 3.71249e+02 3.40461e+02 4.39046e+02 7.36607e+02 8.98245e+02 1.06580e+03 8.00818e+02 8.37289e+02 [18] 8.60849e+02 8.57644e+02 8.68042e+02 8.64769e+02 5.41615e+02 3.79254e+01 2.25170e-09 2.24417e-09 2.24435e-09 2.24318e-09 2.23910e-09 2.23552e-09 1.00000e-02 2.31996e+00 2.00892e+01 7.62612e+01 1.58152e+02 [35] 2.94857e+02 2.43349e+02 3.63265e+02 4.41769e+02 4.09168e+02 3.73260e+02 2.39956e+02 6.09519e+01 3.99840e+01 4.12993e+01 6.38418e+01 8.69061e+01 5.42347e+01 4.91593e+01 4.24806e+01 5.14451e+01 4.73900e+00 [52] 5.53085e+01

$F_fsh_1 [,1] [,2] [,3] [1,] 1970 0.00993833 0.03192130 [2,] 1971 0.01473840 0.04733870 [3,] 1972 0.00674561 0.02166650 [4,] 1973 0.00861514 0.02767130 [5,] 1974 0.01677620 0.05388420 [6,] 1975 0.02086250 0.06700890 [7,] 1976 0.02552690 0.08199070 [8,] 1977 0.02226580 0.07151620 [9,] 1978 0.04398880 0.14128900 [10,] 1979 0.03697100 0.11874800 [11,] 1980 0.02544620 0.08173140 [12,] 1981 0.03757870 0.12070000 [13,] 1982 0.07257400 0.23310300 [14,] 1983 0.02809760 0.09024760 [15,] 1984 0.06546040 0.16717900 [16,] 1985 0.04414540 0.09494920 [17,] 1986 0.00436001 0.00851048 [18,] 1987 0.02067660 0.03020860 [19,] 1988 0.01751130 0.02742440 [20,] 1989 0.01579730 0.03190110 [21,] 1990 0.01427330 0.03148230 [22,] 1991 0.01730280 0.03291990 [23,] 1992 0.01758470 0.03135850 [24,] 1993 0.02054640 0.03689860 [25,] 1994 0.01411460 0.02586340 [26,] 1995 0.01941190 0.04353800 [27,] 1996 0.03525330 0.09510790 [28,] 1997

0.01441390 0.04148010 [29,] 1998 0.00415611 0.01093830 [30,] 1999 0.00654013 0.01601480 [31,] 2000 0.00992589 0.02521300 [32,] 2001 0.01974120 0.05198670 [33,] 2002 0.00840511 0.02173000 [34,] 2003 0.01171490 0.03280430 [35,] 2004 0.01279950 0.03970440 [36,] 2005 0.02023870 0.05975950 [37,] 2006 0.02485360 0.08291080 [38,] 2007 0.03418670 0.10980200 [39,] 2008 0.04682520 0.13118000 [40,] 2009 0.04752940 0.14559600 [41,] 2010 0.06699310 0.17385600 [42,] 2011 0.01098650 0.02238420 [43,] 2012 0.00427719 0.01036380 [44,] 2013 0.00528907 0.00951096 [45,] 2014 0.00576600 0.01010100 [46,] 2015 0.00716261 0.01155430 [47,] 2016 0.00560160 0.00794378 [48,] 2017 0.00819638 0.01030490 [49,] 2018 0.00259274 0.00355169 [50,] 2019 0.00166784 0.00270504 [51,] 2020 0.00455542 0.00812722 [52,] 2021 0.00344886 0.00615304

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$Like_Comp [1] 1.09388e+00 3.53420e+02 4.60395e+02 2.67569e+02 2.64167e+02 1.03692e+02 0.00000e+00 1.33961e+01 1.16895e+00 9.70438e-03 2.69025e-01 0.00000e+00 1.28340e-02 1.46519e+03

$Sel_Fshry_1 [1] 28.80140 28.48970 5.47443 0.00000

$Sel_Fshry_2 [1] 21.88820000 23.27170000 0.00547995 0.00000000

$Sel_Fshry_3 [1] 101.119000 36.283000 0.113957 0.000000

$Sel_Fshry_4 [1] 9.62400000 12.49510000 0.00352046 0.00000000

$Survey_Index_1 [1] 38.8033

$Survey_Index_2 [1] 64.2943

$Survey_Index_3 [1] 16.9313

$Survey_Index_4 [1] 11.3642

$Survey_Index_5 [1] 75.2963

$Survey_Index_6 [1] 46.08

$Survey_Index_7 [1] 11.3977

$Age_Survey_1 [1] 21.24802

$Age_Survey_2 [1] 53.76959

$Age_Survey_3 [1] 0

$Age_Survey_4 [1] 28.67468

$Age_Survey_5 [1] 0

$Age_Survey_6 [1] 0

$Age_Survey_7 [1] 0

$Sel_Survey_1 [1] 4.58451 0.59294 0.00000

$Sel_Survey_2 [1] 1.74703 1.58337 0.00000

$Sel_Survey_3 [1] 0 0 0

$Sel_Survey_4 [1] 2.87840 2.00985 0.00000

$Sel_Survey_5 [1] 0 0 0

$Sel_Survey_6 [1] 0 0 0

$Sel_Survey_7 [1] 0 0 0

$Rec_Pen [1] 0.60000 2.39911 9.90204 -10.21651 -0.91569

$m_sigmar [1] 0.66854

$sigmar [1] 0.6

$rec_dev [,1] [,2] [1,] 1959 -0.43884 [2,] 1960 -0.51187 [3,] 1961 -0.59140 [4,] 1962 -0.67818 [5,] 1963 -0.76981 [6,] 1964 -0.86271 [7,] 1965 -0.83127 [8,] 1966 -0.83676 [9,] 1967 -0.68503 [10,] 1968 -0.62936 [11,] 1969 -0.54822 [12,] 1970 -0.38443 [13,] 1971 -0.42127 [14,] 1972 -0.37525 [15,] 1973 -0.33042 [16,] 1974 -0.13105 [17,] 1975 0.20677 [18,] 1976 0.37772 [19,] 1977 0.34379 [20,] 1978 0.38181 [21,] 1979 0.38432 [22,] 1980 0.42486 [23,] 1981 0.62195 [24,] 1982 0.77479 [25,] 1983 0.58144 [26,] 1984 1.06801 [27,] 1985 1.25869 [28,] 1986 0.66417 [29,] 1987 0.59884 [30,] 1988 0.77421 [31,] 1989 0.63371 [32,] 1990 0.71750 [33,] 1991 0.35071 [34,] 1992 0.38159 [35,] 1993 0.03322 [36,] 1994 -0.03110 [37,] 1995 0.23937 [38,] 1996 0.34582 [39,] 1997 0.64831 [40,] 1998 0.59493 [41,] 1999 0.76041 [42,] 2000 0.79427 [43,] 2001 0.38192 [44,] 2002 -0.00151 [45,] 2003 -0.56603 [46,] 2004 -0.53941 [47,] 2005 -0.78943 [48,] 2006 -0.76469 [49,] 2007 -1.14551 [50,] 2008 -1.25709 [51,] 2009 -0.68249 [52,] 2010 -0.46553 [53,] 2011 -1.04103 [54,] 2012 -0.53852 [55,] 2013 -0.67374 [56,] 2014 -0.39139 [57,] 2015 -0.38829 [58,] 2016 0.03448 [59,] 2017 0.91195 [60,] 2018 0.82000 [61,] 2019 0.29971 [62,] 2020 -0.05452 [63,] 2021 0.03688

$F_Pen [1] 0.0097 0.0000

$Q_Survey_1 [1] 0.18288 0.90245 1.14388 12.00000

$Q_power_Survey_1 [1] 0.18288 1.00000 1.00000 1.20000

$Q_Survey_2 [1] 0.00088 0.04804 0.02905 12.00000

$Q_power_Survey_2 [1] 0.00088 1.00000 1.00000 1.20000

$Q_Survey_3 [1] 0.02201 0.00007 0.00016 12.00000

$Q_power_Survey_3 [1] 0.02201 1.00000 1.00000 1.20000

$Q_Survey_4 [1] 0.00003 0.46991 0.51917 12.00000

$Q_power_Survey_4 [1] 0.00003 1.00000 1.00000 1.20000

$Q_Survey_5 [1] 0.00005 0.00202 0.00228 12.00000

$Q_power_Survey_5 [1] 0.00005 1.00000 1.00000 1.20000

$Q_Survey_6 [1] 0.04545 0.00021 0.00799 12.00000

$Q_power_Survey_6 [1] 0.04545 1.00000 1.00000 1.20000

$Q_Survey_7 [1] 0.01771 0.32424 0.03388 12.00000

$Q_power_Survey_7 [1] 0.01771 1.00000 1.00000 1.20000

$Mest [1] 0.00 0.23 0.23 0.05

$Steep [1] 0.00 0.65 0.65 300.00

$Sigmar [1] 0.0 0.6 0.6 15.0

$Num_parameters_Est [1] 1695

$Steep_Prior [1] 0.65 300.00 -6.00

$sigmarPrior [1] 0.6 15.0 -4.0

$Rec_estimated_in_styr_endyr [1] 1959 2021

$SR_Curve_fit___in_styr_endyr_1 [1] 2000 2018

$Model_styr_endyr [1] 1970 2021

$M_prior [1] 0.23 0.05 -4.00

$qprior [,1] [,2] [,3] [1,] 1.14388 12 3 [2,] 0.02905 12 5 [3,] 0.00016 12 3 [4,] 0.51917 12 3 [5,] 0.00228 12 3 [6,] 0.00799 12 4 [7,] 0.03388 12 4

$q_power_prior [,1] [,2] [,3] [1,] 1 1.2 -1 [2,] 1 1.2 -1 [3,] 1 1.2 -1 [4,] 1 1.2 -1 [5,] 1 1.2 -1 [6,] 1 1.2 -1 [7,] 1 1.2 -1

$cv_catchbiomass [1] 0.05

$Projection_years [1] 20

$Fsh_sel_opt_fish [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [,14] [,15] [,16] [,17] [,18] [,19] [,20] [,21] [,22] [,23] [,24] [,25] [,26] [,27] [,28] [,29] [,30] [,31] [,32] [,33] [,34] [,35] [,36] [1,] 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 [2,] 2 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 [3,] 3 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 [4,] 4 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 [,37] [,38] [,39] [,40] [,41] [,42] [,43] [,44] [,45] [,46] [,47] [,48] [,49] [,50] [,51] [,52] [,53] [,54] [1,] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 [2,] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 [3,] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 [4,] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

$Survey_Sel_Opt_Survey [,1] [,2] [1,] 1 1 [2,] 2 1 [3,] 3 1 [4,] 4 1 [5,] 5 1 [6,] 6 1 [7,] 7 1

$Phase_survey_Sel_Coffs [,1] [1,] 2 [2,] 5 [3,] -5 [4,] 3 [5,] -5 [6,] -5 [7,] -5

$Fshry_Selages [1] 9 10 7 9

$Survy_Selages [1] 10 10 10 10 10 10 10

$Phase_for_age_spec_fishery [1] 2 3 4 3

$Phase_for_logistic_fishery [1] -1 -1 -1 -1

$Phase_for_dble_logistic_fishery [1] -1 -1 -1 -1

$Phase_for_age_spec_survey [1] 2 5 -5 3 -5 -5 -5

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$Phase_for_dble_logistic_indy [1] -1 -1 -1 -1 -1 -1 -1

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0.43400 0.45300 0.59000 1.11500 [10,] 1979 0.05000 0.10800 0.16300 0.17900 0.21700 0.27400 0.37000 0.42000 0.47400 0.62900 0.63300 1.11500 [11,] 1980 0.05000 0.06900 0.11800 0.21000 0.25600 0.32400 0.41000 0.45100 0.51100 0.99800 0.88000 1.11500 [12,] 1981 0.05000 0.09400 0.13900 0.21400 0.26900 0.33100 0.41200 0.48100 0.58000 0.66100 1.11200 1.11500 [13,] 1982 0.07100 0.09300 0.16800 0.20200 0.24800 0.30500 0.35600 0.41100 0.44600 0.47100 0.71900 1.11500 [14,] 1983 0.08400 0.09900 0.11900 0.22100 0.26400 0.31400 0.37700 0.42900 0.47500 0.52800 0.54000 1.11500 [15,] 1984 0.05000 0.16400 0.18600 0.21700 0.27300 0.34500 0.39400 0.43700 0.49700 0.56800 0.78600 1.11500 [16,] 1985 0.05000 0.16700 0.17300 0.22400 0.27100 0.34000 0.40100 0.46500 0.53600 0.58200 0.72600 1.11500 [17,] 1986 0.09600 0.09900 0.14300 0.22200 0.28900 0.33200 0.41800 0.49700 0.55000 0.86900 0.88000 1.11500 [18,] 1987 0.09200 0.12100 0.14600 0.18900 0.23300 0.33600 0.42700 0.47700 0.51300 0.65000 0.80300 1.11500 [19,] 1988 0.05000 0.11000 0.16700 0.19700 0.23000 0.29800 0.47200 0.54500 0.58600 0.60950 0.88000 1.11500 [20,] 1989 0.05000 0.12300 0.16700 0.23000 0.27000 0.31000 0.37900 0.49100 0.54100 0.56900 0.71300 1.11500 [21,] 1990 0.06900 0.09900 0.16000 0.24800 0.29000 0.33800 0.40900 0.53300 0.65100 0.67700 0.75600 1.11500 [22,] 1991 0.04900 0.12100 0.14300 0.20100 0.27700 0.36600 0.40800 0.47800 0.63700 0.72000 0.79400 0.88300 [23,] 1992 0.06900 0.09200 0.12700 0.20100 0.26800 0.30000 0.37300 0.44400 0.51200 0.59500 0.68100 0.78600 [24,] 1993 0.02100 0.11600 0.15200 0.20500 0.29800 0.36400 0.42200 0.48900 0.52800 0.59600 0.77400 0.88900 [25,] 1994 0.05900 0.09700 0.10700 0.23500 0.29100 0.33000 0.38700 0.45900 0.56500 0.74800 0.79800 0.89800 [26,] 1995 0.06900 0.10100 0.13700 0.18600 0.26300 0.32100 0.35700 0.43400 0.56100 0.66800 0.88000 1.11500 [27,] 1996 0.06700 0.00000 0.14000 0.17000 0.22900 0.29500 0.36700 0.50700 0.65700 0.63900 0.88000 1.11500 [28,] 1997 0.02900 0.06300 0.12500 0.17700 0.24600 0.35700 0.50300 0.61500 0.58400 0.72800 0.88000 1.11500 [29,] 1998 0.00000 0.08200 0.10400 0.19500 0.24900 0.29000 0.39000 0.47500 0.63400 0.72800 0.88000 1.11500 [30,] 1999 0.07100 0.07400 0.08900 0.14700 0.27000 0.31500 0.44600 0.72200 0.58400 0.72800 0.88000 1.11500 [31,] 2000 0.04300 0.05400 0.13800 0.19100 0.22500 0.25100 0.37200 0.48800 0.58400 0.72800 0.88000 1.11500 [32,] 2001 0.06600 0.09300 0.11200 0.13300 0.20400 0.28600 0.42100 0.48800 0.58400 0.72800 0.88000 1.11500 [33,] 2002 0.02900 0.05900 0.09200 0.17200 0.23800 0.32700 0.39800 0.41600 0.62800 0.72800 0.88000 1.11500 [34,] 2003 0.03600 0.08200 0.10200 0.14100 0.22700 0.30900 0.41600 0.46400 0.53400 0.72800 0.88000 1.11500 [35,] 2004 0.03700 0.07800 0.16400 0.18600 0.20300 0.25700 0.34200 0.48800 0.58400 0.72800 0.88000 1.11500 [36,] 2005 0.02900 0.07600 0.11100 0.17500 0.22200 0.26800 0.28100 0.48800 0.58400 0.72800 0.88000 1.11500 [37,] 2006 0.03200 0.07400 0.11400 0.13200 0.20400 0.37400 0.44200 0.50600 0.60600 0.72800 0.88000 1.11500 [38,] 2007 0.08700 0.07500 0.12200 0.15800 0.22200 0.29600 0.40400 0.51400 0.61400 0.72300 0.72300 1.11500 [39,] 2008 0.04200 0.04700 0.06600 0.18700 0.24300 0.29100 0.38800 0.56300 0.61600 0.74800 0.88000 1.11500 [40,] 2009 0.01500 0.04700 0.10600 0.13800 0.23900 0.28500 0.33500 0.52600 0.58400 0.72800 0.88000 1.11500 [41,] 2010 0.01300 0.04800 0.10100 0.17200 0.23300 0.30100 0.39700 0.49300 0.63900 0.77200 0.88000 1.11500 [42,] 2011 0.01900 0.06500 0.09500 0.16700 0.27600 0.31400 0.39800 0.48800 0.58400 0.72800 0.88000 1.11500 [43,] 2012 0.01600 0.04800 0.08800 0.20200 0.23500 0.26900 0.39600 0.48800 0.58400 0.72800 0.88000 1.11500 [44,] 2013 0.03800 0.05200 0.06900 0.15100 0.25500 0.43000 0.49500 0.66400 0.52500 0.68700 0.82100 1.08600 [45,] 2014 0.01800 0.04000 0.08200 0.18900 0.24800 0.31300 0.39600 0.48800 0.58400 0.72800 0.88000 1.11500 [46,] 2015 0.02743 0.05809 0.17667 0.18307 0.29804 0.44188 0.62101 0.52015 0.58320 0.72860 0.86820 1.10920 [47,] 2016 0.02743 0.05809 0.15800 0.19485 0.23479 0.30017 0.35293 0.53499 0.69181 0.74158 0.85947 0.97370 [48,] 2017 0.02360 0.06290 0.13987 0.16353 0.18110 0.22345 0.29928 0.39963 0.60029 0.52841 0.88000 1.11500 [49,] 2018 0.01649 0.04083 0.09349 0.19901 0.23470 0.25852 0.27182 0.32268 0.32268 0.52841 0.88000 1.11500 [50,] 2019 0.01301 0.03337 0.12809 0.18049 0.28014 0.44409 0.50107 0.52762 0.73243 0.85293 0.88798 1.51222 [51,] 2020 0.01301 0.03337 0.21456 0.21050 0.24160 0.30405 0.40812 0.47664 0.84163 1.28859 1.75430 2.99875 [52,] 2021 0.01837 0.05221 0.09670 0.39547 0.41634 0.47689 0.55307 0.75616 0.98723 1.22510 1.52496 2.99875

$wt_fsh_2 [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [1,] 1970 0.05200 0.09300 0.13100 0.17800 0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [2,] 1971 0.05200 0.09300 0.13100 0.17800 0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [3,] 1972 0.05200 0.09300 0.13100 0.17800 0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [4,] 1973 0.05200 0.09300 0.13100 0.17800 0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [5,] 1974 0.05200 0.09300 0.13100 0.17800 0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [6,] 1975 0.05200 0.09300 0.13100 0.17800 0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [7,] 1976 0.05200 0.07800 0.15500 0.21400 0.27500 0.33600 0.39400 0.47200 0.63200 0.71400 0.89800 1.53800 [8,] 1977 0.05500 0.09200 0.10900 0.23600 0.27500 0.31400 0.37500 0.45600 0.52100 0.73200 0.65100 1.13700 [9,] 1978 0.05200 0.08400 0.10400 0.14700 0.21100 0.32700 0.39400 0.44900 0.51400 0.58300 0.63100 1.53800 [10,] 1979 0.05200 0.10800 0.16000 0.19900 0.24100 0.30100 0.38800 0.46600 0.58800 0.87100 1.26500 1.97200 [11,] 1980 0.02600 0.06000 0.13200 0.23100 0.27200 0.35000 0.44700 0.51900 0.71600 0.82000 1.07300 1.85400 [12,] 1981 0.05200 0.09500 0.14900 0.24200 0.29400 0.34000 0.40700 0.50300 0.63700 0.76500 1.18400 1.90000 [13,] 1982 0.05500 0.08500 0.16600 0.20700 0.26900 0.32300 0.37800 0.47200 0.53600 0.64400 0.98700 1.18500 [14,] 1983 0.07000 0.09900 0.12200 0.23000 0.27300 0.32000 0.37400 0.46100 0.59600 0.70900 1.19600 1.76900 [15,] 1984 0.03500 0.13500 0.15400 0.18500 0.26600 0.33000 0.38300 0.44900 0.57700 0.68500 1.01200 1.84600 [16,] 1985 0.05800 0.14800 0.18100 0.22300 0.27000 0.33900 0.39800 0.47300 0.57300 0.79600 1.37600 1.64700 [17,] 1986 0.07300 0.07500 0.17200 0.24700 0.28600 0.34600 0.42700 0.51800 0.64000 0.84400 1.35100 2.11000 [18,] 1987 0.07600 0.11700 0.14000 0.19100 0.27000 0.35700 0.43400 0.50300 0.57700 0.68900 1.08900 1.97900 [19,] 1988 0.10000 0.12400 0.15900 0.19700 0.23300 0.34200 0.44400 0.51200 0.58800 0.75000 1.01200 1.37200 [20,] 1989 0.05200 0.10300 0.22000 0.24100 0.27800 0.33900 0.46700 0.58500 0.70200 0.77900 0.88000 1.53800 [21,] 1990 0.06400 0.09100 0.15300 0.26400 0.30900 0.37300 0.46100 0.58200 0.69400 0.83500 0.97000 1.59800 [22,] 1991 0.03700 0.10600 0.13200 0.18600 0.27100 0.38100 0.45100 0.54200 0.66700 0.78700 0.90100 1.05300 [23,] 1992 0.06300 0.08300 0.11800 0.17700 0.23900 0.27500 0.40900 0.52400 0.59400 0.70900 0.85100 1.04600 [24,] 1993 0.01100 0.08900 0.12100 0.18100 0.24600 0.32000 0.40800 0.57900 0.71900 0.85300 0.96500 1.17400 [25,] 1994 0.04100 0.08400 0.11200 0.22400 0.27000 0.33600 0.46200 0.64300 0.80800 0.86800 1.05800 1.42100 [26,] 1995 0.07000 0.09800 0.14500 0.19200 0.27000 0.34000 0.42900 0.57700 0.80700 0.96500 1.11500 1.36700 [27,] 1996 0.06100 0.09200 0.15100 0.19100 0.28000 0.35200 0.52400 0.68300 0.94500 1.21600 1.42600 1.47700 [28,] 1997 0.10400 0.10600 0.14600 0.20100 0.26000 0.35500 0.49500 0.68300 0.88400 1.08800 1.46700 1.64700 [29,] 1998 0.08400 0.12800 0.13800 0.17800 0.24800 0.34000 0.54500 0.80600 1.03500 1.24600 1.41200 1.65500 [30,] 1999 0.09000 0.10900 0.13400 0.17400 0.25000 0.33100 0.46500 0.74200 1.02100 1.25800 1.37600 1.77600 [31,] 2000 0.04300 0.06400 0.16300 0.19600 0.25500 0.34600 0.46600 0.75600 0.99900 1.14100 1.22800 1.56300 [32,] 2001 0.06600 0.09800 0.12200 0.17900 0.25800 0.32500 0.46100 0.61400 0.82800 1.07400 1.36000 1.67100 [33,] 2002 0.03100 0.07400 0.13000 0.20000 0.25700 0.32900 0.44500 0.64500 0.88300 1.10200 1.32100 1.64900 [34,] 2003 0.03600 0.08600 0.11700 0.18600 0.24500 0.30700 0.40000 0.56400 0.76800 1.00500 1.20900 1.53700 [35,] 2004 0.03400 0.08000 0.15800 0.19300 0.24700 0.30700 0.38700 0.52800 0.70000 0.89700 1.08700 1.54100 [36,] 2005 0.02900 0.07500 0.11300 0.19600 0.25900 0.31800 0.39900 0.51700 0.64100 0.76700 0.91800 1.29600 [37,] 2006 0.03300 0.07600 0.11600 0.14100 0.26100 0.35000 0.41900 0.51600 0.63100 0.75200 0.92400 1.26300 [38,] 2007 0.08600 0.07400 0.12100 0.17200 0.22600 0.33100 0.43100 0.51000 0.62100 0.75600 0.90300 1.17700 [39,] 2008 0.03600 0.04800 0.06900 0.18600 0.25400 0.31200 0.41600 0.51500 0.60500 0.71900 0.86100 1.14800 [40,] 2009 0.01400 0.04500 0.10900 0.14200 0.25300 0.33000 0.41100 0.53200 0.62500 0.76400 0.88600 1.14400 [41,] 2010 0.01400 0.05200 0.10100 0.17500 0.23700 0.31300 0.41500 0.53900 0.64900 0.78700 0.96400 1.47300 [42,] 2011 0.01900 0.06700 0.10100 0.19000 0.28700 0.35300 0.46600 0.61300 0.77400 0.92300 1.17300 1.51400 [43,] 2012 0.00700 0.01400 0.08200 0.20200 0.26400 0.35300 0.47600 0.55800 0.71100 0.91200 1.14600 1.60000 [44,] 2013 0.05400 0.15800 0.25100 0.26000 0.31800 0.38500 0.45000 0.55300 0.70500 0.82900 1.11700 1.97700 [45,] 2014 0.05200 0.09300 0.18162 0.24712 0.37482 0.48468 0.53435 0.68241 1.09370 1.28054 1.30202 1.65623 [46,] 2015 0.05116 0.11220 0.19093 0.25619 0.34527 0.42854 0.50445 0.62719 0.93521 1.06949 1.20602 1.44665 [47,] 2016 0.05031 0.13140 0.20024 0.26527 0.31572 0.37239 0.47455 0.57196 0.77672 0.85845 1.11003 1.23707 [48,] 2017 0.01673 0.05839 0.20054 0.23994 0.30340 0.38184 0.46816 0.56217 0.72122 0.95326 1.09571 1.61643 [49,] 2018 0.01555 0.03095 0.21244 0.24110 0.30509 0.37765 0.49450 0.59430 0.77991 0.91597 1.34619 1.82370 [50,] 2019 0.01460 0.02499 0.18746 0.26986 0.30461 0.39068 0.48069 0.57868 0.68549 0.76116 0.89770 1.62786 [51,] 2020 0.01460 0.02499 0.24613 0.23171 0.30291 0.42474 0.55987 0.68489 0.80824 1.00375 1.18741 1.43616 [52,] 2021 0.01460 0.02499 0.09462 0.27236 0.30312 0.34675 0.42703 0.60435 0.71289 0.80864 1.10557 2.12768

$wt_fsh_3 [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [1,] 1970 0.0450 0.1710 0.377 0.6420 0.945 1.265 1.587 1.9000 2.196 2.470 2.7210 2.9460 [2,] 1971 0.0450 0.1710 0.377 0.6430 0.946 1.266 1.588 1.9020 2.198 2.472 2.7230 2.9490 [3,] 1972 0.0300 0.1300 0.306 0.5480 0.835 1.148 1.470 1.7890 2.095 2.382 2.6470 2.8870 [4,] 1973 0.0370 0.1470 0.330 0.5680 0.842 1.134 1.430 1.7180 1.991 2.246 2.4780 2.6880 [5,] 1974 0.0380 0.1470 0.326 0.5580 0.825 1.108 1.393 1.6710 1.934 2.178 2.4020 2.6030 [6,] 1975 0.0340 0.1360 0.310 0.5400 0.808 1.095 1.387 1.6740 1.946 2.201 2.4340 2.6450 [7,] 1976 0.0440 0.1600 0.340 0.5670 0.822 1.087 1.351 1.6060 1.845 2.065 2.2660 2.4460 [8,] 1977 0.0320 0.1300 0.294 0.5100 0.760 1.028 1.300 1.5660 1.818 2.054 2.2700 2.4650 [9,] 1978 0.0320 0.1290 0.295 0.5160 0.774 1.050 1.332 1.6080 1.872 2.117 2.3430 2.5470 [10,] 1979 0.0360 0.1380 0.304 0.5180 0.762 1.020 1.280 1.5320 1.770 1.991 2.1930 2.3750 [11,] 1980 0.0360 0.1360 0.298 0.5060 0.743 0.994 1.245 1.4900 1.721 1.934 2.1300 2.3060 [12,] 1981 0.0410 0.1480 0.314 0.5240 0.758 1.003 1.247 1.4810 1.702 1.905 2.0890 2.2550 [13,] 1982 0.0390 0.1440 0.309 0.5190 0.755 1.002 1.249 1.4880 1.712 1.920 2.1080 2.2780 [14,] 1983 0.0420 0.1380 0.280 0.4510 0.638 0.828 1.014 1.1910 1.356 1.507 1.6430 1.7640 [15,] 1984 0.0440 0.1560 0.328 0.5410 0.778 1.024 1.267 1.5010 1.719 1.921 2.1030 2.2670 [16,] 1985 0.0400 0.1490 0.322 0.5410 0.789 1.048 1.308 1.5580 1.794 2.012 2.2110 2.3890 [17,] 1986 0.0420 0.1510 0.323 0.5390 0.781 1.033 1.285 1.5270 1.755 1.965 2.1560 2.3270 [18,] 1987 0.0340 0.1320 0.294 0.5040 0.745 1.001 1.260 1.5120 1.751 1.973 2.1760 2.3590 [19,] 1988 0.0380 0.1450 0.315 0.5330 0.780 1.041 1.302 1.5540 1.793 2.013 2.2150 2.3960 [20,] 1989 0.0440 0.1580 0.337 0.5610 0.812 1.074 1.334 1.5850 1.821 2.038 2.2360 2.4130 [21,] 1990 0.0420 0.1500 0.320 0.5320 0.769 1.017 1.263 1.4990 1.722 1.927 2.1130 2.2800 [22,] 1991 0.0390 0.1420 0.305 0.5110 0.743 0.985 1.227 1.4610 1.680 1.883 2.0680 2.2340 [23,] 1992 0.0400 0.1480 0.318 0.5340 0.776 1.031 1.286 1.5310 1.763 1.976 2.1710 2.3460 [24,] 1993 0.0390 0.1470 0.323 0.5490 0.807 1.080 1.354 1.6200 1.871 2.104 2.3170 2.5080 [25,] 1994 0.0360 0.1470 0.335 0.5840 0.874 1.186 1.503 1.8130 2.109 2.385 2.6380 2.8670 [26,] 1995 0.0380 0.1460 0.318 0.5400 0.792 1.058 1.325 1.5830 1.827 2.053 2.2600 2.4460 [27,] 1996 0.0380 0.1450 0.317 0.5370 0.788 1.053 1.318 1.5760 1.820 2.045 2.2510 2.4360 [28,] 1997 0.0450 0.1520 0.312 0.5060 0.720 0.940 1.155 1.3610 1.553 1.729 1.8890 2.0310 [29,] 1998 0.0400 0.1400 0.294 0.4830 0.693 0.911 1.126 1.3330 1.526 1.703 1.8640 2.0080 [30,] 1999 0.0370 0.1460 0.324 0.5570 0.824 1.107 1.394 1.6730 1.938 2.183 2.4080 2.6110 [31,] 2000 0.0350 0.1450 0.336 0.5920 0.893 1.218 1.550 1.8770 2.189 2.481 2.7500 2.9940 [32,] 2001 0.0330 0.1390 0.324 0.5720 0.864 1.180 1.504 1.8220 2.127 2.412 2.6740 2.9120 [33,] 2002 0.0360 0.1450 0.330 0.5760 0.861 1.167 1.478 1.7830 2.074 2.344 2.5930 2.8170 [34,] 2003 0.0400 0.1540 0.341 0.5840 0.862 1.157 1.454 1.7430 2.017 2.272 2.5040 2.7140 [35,] 2004 0.0380 0.1490 0.333 0.5740 0.852 1.148 1.447 1.7400 2.017 2.275 2.5110 2.7240 [36,] 2005 0.0370 0.1500 0.341 0.5950 0.890 1.206 1.527 1.8420 2.142 2.422 2.6780 2.9110 [37,] 2006 0.0380 0.1520 0.347 0.6060 0.907 1.230 1.558 1.8800 2.187 2.473 2.7350 2.9730 [38,] 2007 0.0380 0.1490 0.335 0.5790 0.861 1.161 1.465 1.7620 2.044 2.306 2.5460 2.7630 [39,] 2008 0.0360 0.1460 0.334 0.5850 0.876 1.190 1.510 1.8230 2.122 2.400 2.6560 2.8880 [40,] 2009 0.0380 0.1500 0.337 0.5820 0.865 1.167 1.474 1.7730 2.057 2.321 2.5630 2.7820 [41,] 2010 0.0390 0.1500 0.332 0.5670 0.837 1.123 1.411 1.6910 1.956 2.203 2.4280 2.6310 [42,] 2011 0.0310 0.1430 0.351 0.6440 1.000 1.395 1.806 2.2170 2.614 2.990 3.3370 3.6550 [43,] 2012 0.0320 0.1450 0.349 0.6320 0.971 1.344 1.731 2.1150 2.485 2.834 3.1560 3.4490 [44,] 2013 0.0320 0.1450 0.349 0.6320 0.971 1.344 1.731 2.1150 2.485 2.834 3.1560 3.4490 [45,] 2014 0.0320 0.1450 0.349 0.6320 0.971 1.344 1.731 2.1150 2.485 2.834 3.1560 3.4490 [46,] 2015 0.0332 0.1456 0.346 0.6214 0.950 1.310 1.682 2.0506 2.405 2.739 3.0466 3.3266 [47,] 2016 0.0332 0.1456 0.346 0.6214 0.950 1.310 1.682 2.0506 2.405 2.739 3.0466 3.3266 [48,] 2017 0.0332 0.1456 0.346 0.6214 0.950 1.310 1.682 2.0506 2.405 2.739 3.0466 3.3266 [49,] 2018 0.0332 0.1456 0.346 0.6214 0.950 1.310 1.682 2.0506 2.405 2.739 3.0466 3.3266 [50,] 2019 0.0332 0.1456 0.346 0.6214 0.950 1.310 1.682 2.0506 2.405 2.739 3.0466 3.3266 [51,] 2020 0.0332 0.1456 0.346 0.6214 0.950 1.310 1.682 2.0506 2.405 2.739 3.0466 3.3266 [52,] 2021 0.0332 0.1456 0.346 0.6214 0.950 1.310 1.682 2.0506 2.405 2.739 3.0466 3.3266

$wt_fsh_4 [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [1,] 1970 0.05200 0.09300 0.13100 0.17800 0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [2,] 1971 0.05200 0.09300 0.13100 0.17800 0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [3,] 1972 0.05200 0.09300 0.13100 0.17800 0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [4,] 1973 0.05200 0.09300 0.13100 0.17800 0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [5,] 1974 0.05200 0.09300 0.13100 0.17800 0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [6,] 1975 0.05200 0.09300 0.13100 0.17800 0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [7,] 1976 0.05200 0.07800 0.15500 0.21400 0.27500 0.33600 0.39400 0.47200 0.63200 0.71400 0.89800 1.53800 [8,] 1977 0.05500 0.09200 0.10900 0.23600 0.27500 0.31400 0.37500 0.45600 0.52100 0.73200 0.65100 1.13700 [9,] 1978 0.05200 0.08400 0.10400 0.14700

0.21100 0.32700 0.39400 0.44900 0.51400 0.58300 0.63100 1.53800 [10,] 1979 0.05200 0.10800 0.16000 0.19900
0.24100 0.30100 0.38800 0.46600 0.58800 0.87100 1.26500 1.97200 [11,] 1980 0.02600 0.06000 0.13200 0.23100
0.27200 0.35000 0.44700 0.51900 0.71600 0.82000 1.07300 1.85400 [12,] 1981 0.05200 0.09500 0.14900 0.24200
0.29400 0.34000 0.40700 0.50300 0.63700 0.76500 1.18400 1.90000 [13,] 1982 0.05500 0.08500 0.16600 0.20700
0.26900 0.32300 0.37800 0.47200 0.53600 0.64400 0.98700 1.18500 [14,] 1983 0.07000 0.09900 0.12200 0.23000
0.27300 0.32000 0.37400 0.46100 0.59600 0.70900 1.19600 1.76900 [15,] 1984 0.03500 0.13500 0.15400 0.18500
0.26600 0.33000 0.38300 0.44900 0.57700 0.68500 1.01200 1.84600 [16,] 1985 0.05800 0.14800 0.18100 0.22300
0.27000 0.33900 0.39800 0.47300 0.57300 0.79600 1.37600 1.64700 [17,] 1986 0.07300 0.07500 0.17200 0.24700
0.28600 0.34600 0.42700 0.51800 0.64000 0.84400 1.35100 2.11000 [18,] 1987 0.07600 0.11700 0.14000 0.19100
0.27000 0.35700 0.43400 0.50300 0.57700 0.68900 1.08900 1.97900 [19,] 1988 0.10000 0.12400 0.15900 0.19700
0.23300 0.34200 0.44400 0.51200 0.58800 0.75000 1.01200 1.37200 [20,] 1989 0.05200 0.10300 0.22000 0.24100
0.27800 0.33900 0.46700 0.58500 0.70200 0.77900 0.88000 1.53800 [21,] 1990 0.06400 0.09100 0.15300 0.26400
0.30900 0.37300 0.46100 0.58200 0.69400 0.83500 0.97000 1.59800 [22,] 1991 0.03700 0.10600 0.13200 0.18600
0.27100 0.38100 0.45100 0.54200 0.66700 0.78700 0.90100 1.05300 [23,] 1992 0.06300 0.08300 0.11800 0.17700
0.23900 0.27500 0.40900 0.52400 0.59400 0.70900 0.85100 1.04600 [24,] 1993 0.01100 0.08900 0.12100 0.18100
0.24600 0.32000 0.40800 0.57900 0.71900 0.85300 0.96500 1.17400 [25,] 1994 0.04100 0.08400 0.11200 0.22400
0.27000 0.33600 0.46200 0.64300 0.80800 0.86800 1.05800 1.42100 [26,] 1995 0.07000 0.09800 0.14500 0.19200
0.27000 0.34000 0.42900 0.57700 0.80700 0.96500 1.11500 1.36700 [27,] 1996 0.06100 0.09200 0.15100 0.19100
0.28000 0.35200 0.52400 0.68300 0.94500 1.21600 1.42600 1.47700 [28,] 1997 0.10400 0.10600 0.14600 0.20100
0.26000 0.35500 0.49500 0.68300 0.88400 1.08800 1.46700 1.64700 [29,] 1998 0.08400 0.12800 0.13800 0.17800
0.24800 0.34000 0.54500 0.80600 1.03500 1.24600 1.41200 1.65500 [30,] 1999 0.09000 0.10900 0.13400 0.17400
0.25000 0.33100 0.46500 0.74200 1.02100 1.25800 1.37600 1.77600 [31,] 2000 0.04300 0.06400 0.16300 0.19600
0.25500 0.34600 0.46600 0.75600 0.99900 1.14100 1.22800 1.56300 [32,] 2001 0.06600 0.09800 0.12200 0.17900
0.25800 0.32500 0.46100 0.61400 0.82800 1.07400 1.36000 1.67100 [33,] 2002 0.03100 0.07400 0.13000 0.20000
0.25700 0.32900 0.44500 0.64500 0.88300 1.10200 1.32100 1.64900 [34,] 2003 0.03600 0.08600 0.11700 0.18600
0.24500 0.30700 0.40000 0.56400 0.76800 1.00500 1.20900 1.53700 [35,] 2004 0.03400 0.08000 0.15800 0.19300
0.24700 0.30700 0.38700 0.52800 0.70000 0.89700 1.08700 1.54100 [36,] 2005 0.02900 0.07500 0.11300 0.19600
0.25900 0.31800 0.39900 0.51700 0.64100 0.76700 0.91800 1.29600 [37,] 2006 0.03300 0.07600 0.11600 0.14100
0.26100 0.35000 0.41900 0.51600 0.63100 0.75200 0.92400 1.26300 [38,] 2007 0.08600 0.07400 0.12100 0.17200
0.22600 0.33100 0.43100 0.51000 0.62100 0.75600 0.90300 1.17700 [39,] 2008 0.03600 0.04800 0.06900 0.18600
0.25400 0.31200 0.41600 0.51500 0.60500 0.71900 0.86100 1.14800 [40,] 2009 0.01400 0.04500 0.10900 0.14200
0.25300 0.33000 0.41100 0.53200 0.62500 0.76400 0.88600 1.14400 [41,] 2010 0.01400 0.05200 0.10100 0.17500
0.23700 0.31300 0.41500 0.53900 0.64900 0.78700 0.96400 1.47300 [42,] 2011 0.01900 0.06700 0.10100 0.19000
0.28700 0.35300 0.46600 0.61300 0.77400 0.92300 1.17300 1.51400 [43,] 2012 0.00700 0.01400 0.08200 0.20200
0.26400 0.35300 0.47600 0.55800 0.71100 0.91200 1.14600 1.60000 [44,] 2013 0.05200 0.12500 0.26800 0.26300
0.31000 0.36200 0.43100 0.50700 0.67800 0.72600 0.93600 1.14300 [45,] 2014 0.05200 0.09300 0.21714 0.26648
0.37160 0.46986 0.60255 0.64963 0.74736 0.75274 1.63556 1.72040 [46,] 2015 0.05116 0.11269 0.21576 0.27687
0.35856 0.42755 0.54448 0.63244 0.77686 0.83326 1.36493 1.45751 [47,] 2016 0.05031 0.13238 0.21439 0.28726
0.34551 0.38524 0.48642 0.61525 0.80636 0.91378 1.09431 1.19461 [48,] 2017 0.05621 0.09380 0.44480 0.35273
0.36920 0.43705 0.52466 0.61555 0.65331 0.83729 1.07100 1.11000 [49,] 2018 0.05284 0.10639 0.29211 0.30216
0.36210 0.43072 0.53787 0.62681 0.73568 0.83460 1.26696 1.34167 [50,] 2019 0.12506 0.13132 0.19282 0.41566
0.45122 0.61747 0.74028 0.80397 0.96805 1.03429 1.13205 1.16435 [51,] 2020 0.13184 0.19589 0.21236 0.22292
0.50144 0.67483 0.70558 0.90152 1.06666 1.09025 1.22443 1.16435 [52,] 2021 0.13184 0.19589 0.21236 0.22292
0.50144 0.67483 0.70558 0.90152 1.06666 1.09025 1.22443 1.16435

$wt_ind_1 [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [1,] 1970 0.06374 0.12726 0.16427 0.20672
0.26582 0.35811 0.42997 0.52825 0.63551 0.77912 0.89601 1.02151 [2,] 1971 0.06374 0.12726 0.16427 0.20672
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0.26582 0.35811 0.42997 0.52825 0.63551 0.77912 0.89601 1.02151 [18,] 1987 0.06374 0.12726 0.16427 0.20672
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0.26582 0.35811 0.42997 0.52825 0.63551 0.77912 0.89601 1.02151 [25,] 1994 0.06374 0.12726 0.16427 0.20672
0.26582 0.35811 0.42997 0.52825 0.63551 0.77912 0.89601 1.02151 [26,] 1995 0.06374 0.12726 0.16427 0.20672
0.26582 0.35811 0.42997 0.52825 0.63551 0.77912 0.89601 1.02151 [27,] 1996 0.06374 0.12726 0.16427 0.20672
0.26582 0.35811 0.42997 0.52825 0.63551 0.77912 0.89601 1.02151 [28,] 1997 0.06374 0.13935 0.16023 0.19720
0.25054 0.31092 0.39626 0.53576 0.71140 0.89481 1.06571 1.24082 [29,] 1998 0.06374 0.04471 0.15051 0.16825
0.21160 0.28674 0.37876 0.56790 0.81407 1.04635 1.23868 1.42387 [30,] 1999 0.06374 0.09654 0.12132 0.15106
0.18478 0.26382 0.35046 0.48861 0.77170 1.06907 1.30825 1.42945 [31,] 2000 0.06374 0.10990 0.12665 0.17081
0.19786 0.26060 0.35109 0.46986 0.76926 1.01959 1.16200 1.25424 [32,] 2001 0.06374 0.12726 0.14031 0.15858
0.20011 0.25664 0.32327 0.46077 0.61423 0.83119 1.08105 1.37129 [33,] 2002 0.06545 0.12879 0.17378 0.22501
0.28427 0.44366 0.52526 0.61986 0.68187 0.89963 0.80575 1.01677 [34,] 2003 0.06202 0.14310 0.17787 0.23386
0.28890 0.36353 0.46544 0.54541 0.62739 0.71105 0.97689 0.95009 [35,] 2004 0.06374 0.12726 0.20276 0.24570
0.28551 0.36474 0.46449 0.59188 0.67254 0.80846 1.03927 1.11868 [36,] 2005 0.06374 0.12726 0.16427 0.20672
0.20276 0.24565 0.28418 0.36333 0.46604 0.59188 0.67254 0.80846 [37,] 2006 0.06374 0.12726 0.16427 0.26110
0.36595 0.41380 0.49951 0.58869 0.73758 0.87189 0.90558 1.09127 [38,] 2007 0.06374 0.12726 0.16427 0.20672
0.39678 0.42855 0.50709 0.59906 0.67086 0.88883 1.02400 1.14734 [39,] 2008 0.06374 0.12726 0.16427 0.20672
0.26582 0.44584 0.52341 0.59010 0.67824 0.78316 0.86792 0.97414 [40,] 2009 0.06374 0.12726 0.16427 0.20672
0.26582 0.35811 0.42997 0.44584 0.52341 0.59010 0.67824 0.78316 [41,] 2010 0.06374 0.12726 0.16427 0.15196
0.17025 0.35811 0.35594 0.47746 0.57283 0.64733 0.75976 0.86135 [42,] 2011 0.06374 0.12726 0.16427 0.20672
0.26582 0.35811 0.42997 0.58675 0.61185 0.70635 0.77914 0.88128 [43,] 2012 0.06374 0.12726 0.16427 0.20672
0.26582 0.35811 0.42997 0.52825 0.63551 0.77912 0.89601 1.02151 [44,] 2013 0.06374 0.12726 0.16427 0.20672
0.26582 0.35811 0.42997 0.52825 0.63551 0.77912 0.89601 1.02151 [45,] 2014 0.06374 0.12726 0.16427 0.20672
0.26582 0.35811 0.42997 0.52825 0.63551 0.77912 0.89601 1.02151 [46,] 2015 0.06374 0.12726 0.16427 0.20672
0.26582 0.35811 0.42997 0.52825 0.63551 0.77912 0.89601 1.02151 [47,] 2016 0.06374 0.12726 0.16427 0.20672
0.26582 0.35811 0.42997 0.52825 0.63551 0.77912 0.89601 1.02151 [48,] 2017 0.06374 0.12726 0.16427 0.20672
0.26582 0.35811 0.42997 0.52825 0.63551 0.77912 0.89601 1.02151 [49,] 2018 0.06374 0.12726 0.16427 0.20672
0.26582 0.35811 0.42997 0.52825 0.63551 0.77912 0.89601 1.02151 [50,] 2019 0.06374 0.12726 0.16427 0.20672
0.26582 0.35811 0.42997 0.52825 0.63551 0.77912 0.89601 1.02151 [51,] 2020 0.06374 0.12726 0.16427 0.20672
0.26582 0.35811 0.42997 0.52825 0.63551 0.77912 0.89601 1.02151 [52,] 2021 0.06374 0.12726 0.16427 0.20672
0.26582 0.35811 0.42997 0.52825 0.63551 0.77912 0.89601 1.02151

$wt_ind_2 [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [1,] 1970 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [2,] 1971 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [3,] 1972 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [4,] 1973 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [5,] 1974 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [6,] 1975 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [7,] 1976 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [8,] 1977 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [9,] 1978 0.03810 0.08090 0.12270 0.18220

0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [10,] 1979 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [11,] 1980 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [12,] 1981 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [13,] 1982 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [14,] 1983 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [15,] 1984 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [16,] 1985 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [17,] 1986 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [18,] 1987 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [19,] 1988 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [20,] 1989 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [21,] 1990 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [22,] 1991 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [23,] 1992 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [24,] 1993 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [25,] 1994 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [26,] 1995 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [27,] 1996 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [28,] 1997 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [29,] 1998 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [30,] 1999 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [31,] 2000 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [32,] 2001 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [33,] 2002 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [34,] 2003 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [35,] 2004 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [36,] 2005 0.03810 0.08090 0.12270 0.18220
0.24870 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [37,] 2006 0.03860 0.09230 0.13980 0.18240
0.23570 0.30690 0.40710 0.48790 0.63550 0.77910 0.89600 1.02150 [38,] 2007 0.05720 0.10540 0.15180 0.20600
0.22550 0.30940 0.36170 0.48790 0.63550 0.77910 0.89600 1.02150 [39,] 2008 0.05060 0.06990 0.07520 0.09900
0.23990 0.27950 0.40750 0.48790 0.63550 0.77910 0.89600 1.02150 [40,] 2009 0.03810 0.10570 0.14460 0.20260
0.28130 0.32010 0.45300 0.48790 0.63550 0.77910 0.89600 1.02150 [41,] 2010 0.03810 0.04800 0.14260 0.17300
0.23560 0.28540 0.39540 0.48790 0.63550 0.77910 0.89600 1.02150 [42,] 2011 0.03810 0.07470 0.11010 0.17960
0.22400 0.30100 0.33780 0.48790 0.63550 0.77910 0.89600 1.02150 [43,] 2012 0.00600 0.07000 0.09500 0.23300
0.29900 0.34600 0.48700 0.48790 0.63550 0.77910 0.89600 1.02150 [44,] 2013 0.02119 0.02813 0.13370 0.20195
0.30217 0.42805 0.51715 0.56597 0.63551 0.77912 0.89601 1.02151 [45,] 2014 0.03195 0.04462 0.10379 0.16698
0.33186 0.63016 0.63230 0.48788 0.63551 0.77912 0.89601 1.02151 [46,] 2015 0.01987 0.03994 0.13657 0.17591
0.33091 0.45466 0.58032 0.48788 0.63551 0.77912 0.89601 1.02151 [47,] 2016 0.03900 0.06600 0.15300 0.20600
0.28300 0.31300 0.62000 0.65000 0.77600 0.77912 0.74200 1.02151 [48,] 2017 0.02505 0.03421 0.13434 0.18243
0.28377 0.29036 0.60611 0.65000 0.90506 0.77910 0.89600 1.02150 [49,] 2018 0.02600 0.03800 0.13700 0.15000
0.20200 0.30400 0.38600 0.44000 0.83600 1.10000 0.89600 1.02150 [50,] 2019 0.01951 0.03619 0.14831 0.18465
0.24944 0.27837 0.44178 0.61316 0.72793 1.10000 0.89600 1.02150 [51,] 2020 0.03176 0.04759 0.07225 0.20461
0.23314 0.24239 0.26205 0.37580 0.72793 1.10000 0.89600 1.02150 [52,] 2021 0.02533 0.03766 0.07198 0.24191
0.32860 0.32726 0.38975 0.57104 0.48339 1.10000 0.90672 1.02150

$wt_ind_3 [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [1,] 1970 0.05200 0.09300 0.13100 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [2,] 1971 0.05200 0.09300 0.13100 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [3,] 1972 0.05200 0.09300 0.13100 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [4,] 1973 0.05200 0.09300 0.13100 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [5,] 1974 0.05200 0.09300 0.13100 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [6,] 1975 0.05200 0.09300 0.13100 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [7,] 1976 0.05200 0.07800 0.15500 0.21400
0.27500 0.33600 0.39400 0.47200 0.63200 0.71400 0.89800 1.53800 [8,] 1977 0.05500 0.09200 0.10900 0.23600
0.27500 0.31400 0.37500 0.45600 0.52100 0.73200 0.65100 1.13700 [9,] 1978 0.05200 0.08400 0.10400 0.14700

0.21100 0.32700 0.39400 0.44900 0.51400 0.58300 0.63100 1.53800 [10,] 1979 0.05200 0.10800 0.16000 0.19900
0.24100 0.30100 0.38800 0.46600 0.58800 0.87100 1.26500 1.97200 [11,] 1980 0.02600 0.06000 0.13200 0.23100
0.27200 0.35000 0.44700 0.51900 0.71600 0.82000 1.07300 1.85400 [12,] 1981 0.05200 0.09500 0.14900 0.24200
0.29400 0.34000 0.40700 0.50300 0.63700 0.76500 1.18400 1.90000 [13,] 1982 0.05500 0.08500 0.16600 0.20700
0.26900 0.32300 0.37800 0.47200 0.53600 0.64400 0.98700 1.18500 [14,] 1983 0.07000 0.09900 0.12200 0.23000
0.27300 0.32000 0.37400 0.46100 0.59600 0.70900 1.19600 1.76900 [15,] 1984 0.03500 0.13500 0.15400 0.18500
0.26600 0.33000 0.38300 0.44900 0.57700 0.68500 1.01200 1.84600 [16,] 1985 0.05800 0.14800 0.18100 0.22300
0.27000 0.33900 0.39800 0.47300 0.57300 0.79600 1.37600 1.64700 [17,] 1986 0.07300 0.07500 0.17200 0.24700
0.28600 0.34600 0.42700 0.51800 0.64000 0.84400 1.35100 2.11000 [18,] 1987 0.07600 0.11700 0.14000 0.19100
0.27000 0.35700 0.43400 0.50300 0.57700 0.68900 1.08900 1.97900 [19,] 1988 0.10000 0.12400 0.15900 0.19700
0.23300 0.34200 0.44400 0.51200 0.58800 0.75000 1.01200 1.37200 [20,] 1989 0.05200 0.10300 0.22000 0.24100
0.27800 0.33900 0.46700 0.58500 0.70200 0.77900 0.88000 1.53800 [21,] 1990 0.06400 0.09100 0.15300 0.26400
0.30900 0.37300 0.46100 0.58200 0.69400 0.83500 0.97000 1.59800 [22,] 1991 0.03700 0.10600 0.13200 0.18600
0.27100 0.38100 0.45100 0.54200 0.66700 0.78700 0.90100 1.05300 [23,] 1992 0.06300 0.08300 0.11800 0.17700
0.23900 0.27500 0.40900 0.52400 0.59400 0.70900 0.85100 1.04600 [24,] 1993 0.01100 0.08900 0.12100 0.18100
0.24600 0.32000 0.40800 0.57900 0.71900 0.85300 0.96500 1.17400 [25,] 1994 0.04100 0.08400 0.11200 0.22400
0.27000 0.33600 0.46200 0.64300 0.80800 0.86800 1.05800 1.42100 [26,] 1995 0.07000 0.09800 0.14500 0.19200
0.27000 0.34000 0.42900 0.57700 0.80700 0.96500 1.11500 1.36700 [27,] 1996 0.06100 0.09200 0.15100 0.19100
0.28000 0.35200 0.52400 0.68300 0.94500 1.21600 1.42600 1.47700 [28,] 1997 0.10400 0.10600 0.14600 0.20100
0.26000 0.35500 0.49500 0.68300 0.88400 1.08800 1.46700 1.64700 [29,] 1998 0.08400 0.12800 0.13800 0.17800
0.24800 0.34000 0.54500 0.80600 1.03500 1.24600 1.41200 1.65500 [30,] 1999 0.09000 0.10900 0.13400 0.17400
0.25000 0.33100 0.46500 0.74200 1.02100 1.25800 1.37600 1.77600 [31,] 2000 0.04300 0.06400 0.16300 0.19600
0.25500 0.34600 0.46600 0.75600 0.99900 1.14100 1.22800 1.56300 [32,] 2001 0.06600 0.09800 0.12200 0.17900
0.25800 0.32500 0.46100 0.61400 0.82800 1.07400 1.36000 1.67100 [33,] 2002 0.03100 0.07400 0.13000 0.20000
0.25700 0.32900 0.44500 0.64500 0.88300 1.10200 1.32100 1.64900 [34,] 2003 0.03600 0.08600 0.11700 0.18600
0.24500 0.30700 0.40000 0.56400 0.76800 1.00500 1.20900 1.53700 [35,] 2004 0.03400 0.08000 0.15800 0.19300
0.24700 0.30700 0.38700 0.52800 0.70000 0.89700 1.08700 1.54100 [36,] 2005 0.02900 0.07500 0.11300 0.19600
0.25900 0.31800 0.39900 0.51700 0.64100 0.76700 0.91800 1.29600 [37,] 2006 0.03300 0.07600 0.11600 0.14100
0.26100 0.35000 0.41900 0.51600 0.63100 0.75200 0.92400 1.26300 [38,] 2007 0.08600 0.07400 0.12100 0.17200
0.22600 0.33100 0.43100 0.51000 0.62100 0.75600 0.90300 1.17700 [39,] 2008 0.03600 0.04800 0.06900 0.18600
0.25400 0.31200 0.41600 0.51500 0.60500 0.71900 0.86100 1.14800 [40,] 2009 0.01400 0.04500 0.10900 0.14200
0.25300 0.33000 0.41100 0.53200 0.62500 0.76400 0.88600 1.14400 [41,] 2010 0.01400 0.05200 0.10100 0.17500
0.23700 0.31300 0.41500 0.53900 0.64900 0.78700 0.96400 1.47300 [42,] 2011 0.01900 0.06700 0.10100 0.19000
0.28700 0.35300 0.46600 0.61300 0.77400 0.92300 1.17300 1.51400 [43,] 2012 0.00700 0.01400 0.08200 0.20200
0.26400 0.35300 0.47600 0.55800 0.71100 0.91200 1.14600 1.60000 [44,] 2013 0.05400 0.15800 0.25100 0.26000
0.31800 0.38500 0.45000 0.55300 0.70500 0.82900 1.11700 1.97700 [45,] 2014 0.05200 0.09300 0.18162 0.24712
0.37482 0.48468 0.53435 0.68241 1.09370 1.28054 1.30202 1.65623 [46,] 2015 0.05116 0.11220 0.19093 0.25619
0.34527 0.42854 0.50445 0.62719 0.93521 1.06949 1.20602 1.44665 [47,] 2016 0.05031 0.13140 0.20024 0.26527
0.31572 0.37239 0.47455 0.57196 0.77672 0.85845 1.11003 1.23707 [48,] 2017 0.01673 0.05839 0.20054 0.23994
0.30340 0.38184 0.46816 0.56217 0.72122 0.95326 1.09571 1.61643 [49,] 2018 0.01555 0.03095 0.21244 0.24110
0.30509 0.37765 0.49450 0.59430 0.77991 0.91597 1.34619 1.82370 [50,] 2019 0.01460 0.02499 0.18746 0.26986
0.30461 0.39068 0.48069 0.57868 0.68549 0.76116 0.89770 1.62786 [51,] 2020 0.01460 0.02499 0.24613 0.23171
0.30291 0.42474 0.55987 0.68489 0.80824 1.00375 1.18741 1.43616 [52,] 2021 0.01460 0.02499 0.09462 0.27236
0.30312 0.34675 0.42703 0.60435 0.71289 0.80864 1.10557 2.12768

$wt_ind_4 [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [1,] 1970 0.0520 0.0930 0.1310 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [2,] 1971 0.0520 0.0930 0.1310 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [3,] 1972 0.0520 0.0930 0.1310 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [4,] 1973 0.0520 0.0930 0.1310 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [5,] 1974 0.0520 0.0930 0.1310 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [6,] 1975 0.0520 0.0930 0.1310 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [7,] 1976 0.0520 0.0780 0.1550 0.21400
0.27500 0.33600 0.39400 0.47200 0.63200 0.71400 0.89800 1.53800 [8,] 1977 0.0547 0.0920 0.1090 0.23600
0.27500 0.31400 0.37500 0.45600 0.52100 0.73200 0.65100 1.13700 [9,] 1978 0.0520 0.0840 0.1040 0.14700

0.21100 0.32700 0.39400 0.44900 0.51400 0.58300 0.63100 1.53800 [10,] 1979 0.0520 0.1080 0.1600 0.19900
0.24100 0.30100 0.38800 0.46600 0.58800 0.87100 1.26500 1.97200 [11,] 1980 0.0257 0.0600 0.1320 0.23100
0.27200 0.35000 0.44700 0.51900 0.71600 0.82000 1.07300 1.85400 [12,] 1981 0.0520 0.0950 0.1490 0.24200
0.29400 0.34000 0.40700 0.50300 0.63700 0.76500 1.18400 1.90000 [13,] 1982 0.0552 0.0850 0.1660 0.20700
0.26900 0.32300 0.37800 0.47200 0.53600 0.64400 0.98700 1.18500 [14,] 1983 0.0704 0.0990 0.1220 0.23000
0.27300 0.32000 0.37400 0.46100 0.59600 0.70900 1.19600 1.76900 [15,] 1984 0.0346 0.1350 0.1540 0.18500
0.26600 0.33000 0.38300 0.44900 0.57700 0.68500 1.01200 1.84600 [16,] 1985 0.0575 0.1480 0.1810 0.22300
0.27000 0.33900 0.39800 0.47300 0.57300 0.79600 1.37600 1.64700 [17,] 1986 0.0727 0.0750 0.1720 0.24700
0.28600 0.34600 0.42700 0.51800 0.64000 0.84400 1.35100 2.11000 [18,] 1987 0.0760 0.1170 0.1400 0.19100
0.27000 0.35700 0.43400 0.50300 0.57700 0.68900 1.08900 1.97900 [19,] 1988 0.1003 0.1240 0.1590 0.19700
0.23300 0.34200 0.44400 0.51200 0.58800 0.75000 1.01200 1.37200 [20,] 1989 0.0520 0.1030 0.2200 0.24100
0.27800 0.33900 0.46700 0.58500 0.70200 0.77900 0.88000 1.53800 [21,] 1990 0.0644 0.0910 0.1530 0.26400
0.30900 0.37300 0.46100 0.58200 0.69400 0.83500 0.97000 1.59800 [22,] 1991 0.0372 0.1060 0.1320 0.18600
0.27100 0.38100 0.45100 0.54200 0.66700 0.78700 0.90100 1.05300 [23,] 1992 0.0626 0.0830 0.1180 0.17700
0.23900 0.27500 0.40900 0.52400 0.59400 0.70900 0.85100 1.04600 [24,] 1993 0.0110 0.0890 0.1210 0.18100
0.24600 0.32000 0.40800 0.57900 0.71900 0.85300 0.96500 1.17400 [25,] 1994 0.0412 0.0840 0.1120 0.22400
0.27000 0.33600 0.46200 0.64300 0.80800 0.86800 1.05800 1.42100 [26,] 1995 0.0700 0.0980 0.1450 0.19200
0.27000 0.34000 0.42900 0.57700 0.80700 0.96500 1.11500 1.36700 [27,] 1996 0.0612 0.0920 0.1510 0.19100
0.28000 0.35200 0.52400 0.68300 0.94500 1.21600 1.42600 1.47700 [28,] 1997 0.1044 0.1060 0.1460 0.20100
0.26000 0.35500 0.49500 0.68300 0.88400 1.08800 1.46700 1.64700 [29,] 1998 0.0841 0.1280 0.1380 0.17800
0.24800 0.34000 0.54500 0.80600 1.03500 1.24600 1.41200 1.65500 [30,] 1999 0.0895 0.1090 0.1340 0.17400
0.25000 0.33100 0.46500 0.74200 1.02100 1.25800 1.37600 1.77600 [31,] 2000 0.0426 0.0640 0.1630 0.19600
0.25500 0.34600 0.46600 0.75600 0.99900 1.14100 1.22800 1.56300 [32,] 2001 0.0656 0.0980 0.1220 0.17900
0.25800 0.32500 0.46100 0.61400 0.82800 1.07400 1.36000 1.67100 [33,] 2002 0.0309 0.0740 0.1300 0.20000
0.25700 0.32900 0.44500 0.64500 0.88300 1.10200 1.32100 1.64900 [34,] 2003 0.0358 0.0860 0.1170 0.18600
0.24500 0.30700 0.40000 0.56400 0.76800 1.00500 1.20900 1.53700 [35,] 2004 0.0341 0.0800 0.1580 0.19300
0.24700 0.30700 0.38700 0.52800 0.70000 0.89700 1.08700 1.54100 [36,] 2005 0.0291 0.0750 0.1130 0.19600
0.25900 0.31800 0.39900 0.51700 0.64100 0.76700 0.91800 1.29600 [37,] 2006 0.0328 0.0760 0.1160 0.14100
0.26100 0.35000 0.41900 0.51600 0.63100 0.75200 0.92400 1.26300 [38,] 2007 0.0863 0.0740 0.1210 0.17200
0.22600 0.33100 0.43100 0.51000 0.62100 0.75600 0.90300 1.17700 [39,] 2008 0.0358 0.0483 0.0688 0.18580
0.25380 0.31240 0.41560 0.51520 0.60530 0.71920 0.86130 1.14800 [40,] 2009 0.0138 0.0453 0.1090 0.14230
0.25300 0.32990 0.41120 0.53160 0.62500 0.76430 0.88600 1.14400 [41,] 2010 0.0138 0.0521 0.1013 0.17470
0.23650 0.31330 0.41490 0.53860 0.64880 0.78730 0.96350 1.47300 [42,] 2011 0.0193 0.0667 0.1006 0.19010
0.28720 0.35320 0.46640 0.61300 0.77350 0.92260 1.17330 1.51400 [43,] 2012 0.0073 0.0140 0.0815 0.20180
0.26350 0.35280 0.47580 0.55790 0.71050 0.91170 1.14620 1.60000 [44,] 2013 0.0520 0.1254 0.2682 0.26314
0.30996 0.36198 0.43108 0.50683 0.67832 0.72566 0.93556 1.14265 [45,] 2014 0.0520 0.0930 0.1310 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [46,] 2015 0.0520 0.0930 0.1310 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [47,] 2016 0.0520 0.0930 0.1310 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [48,] 2017 0.0520 0.0930 0.1310 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [49,] 2018 0.0520 0.0930 0.1310 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [50,] 2019 0.0520 0.0930 0.1310 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [51,] 2020 0.0520 0.0930 0.1310 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300 [52,] 2021 0.0520 0.0930 0.1310 0.17800
0.26200 0.29400 0.34000 0.39600 0.54900 0.73800 0.98400 1.09300

$wt_ind_5 [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [1,] 1970 0.04470 0.17124 0.37696 0.64225
0.94486 1.26503 1.58717 1.89996 1.89996 1.89996 1.89996 1.89996 [2,] 1971 0.04472 0.17135 0.37723 0.64273
0.94559 1.26603 1.58845 1.90152 1.90152 1.90152 1.90152 1.90152 [3,] 1972 0.02991 0.12951 0.30634 0.54787
0.83485 1.14783 1.47018 1.78897 1.78897 1.78897 1.78897 1.78897 [4,] 1973 0.03740 0.14733 0.32964 0.56785
0.84208 1.13424 1.42975 1.71788 1.71788 1.71788 1.71788 1.71788 [5,] 1974 0.03774 0.14672 0.32574 0.55819
0.82462 1.10752 1.39295 1.67070 1.67070 1.67070 1.67070 1.67070 [6,] 1975 0.03352 0.13604 0.30980 0.54004
0.80776 1.09509 1.38740 1.67369 1.67369 1.67369 1.67369 1.67369 [7,] 1976 0.04400 0.15961 0.34026 0.56728
0.82155 1.08695 1.35120 1.60568 1.60568 1.60568 1.60568 1.60568 [8,] 1977 0.03235 0.12984 0.29372 0.50974
0.75999 1.02784 1.29973 1.56558 1.56558 1.56558 1.56558 1.56558 [9,] 1978 0.03166 0.12928 0.29549 0.51636

0.77373 1.05038 1.33216 1.60839 1.60839 1.60839 1.60839 1.60839 [10,] 1979 0.03614 0.13833 0.30431 0.51825
0.76220 1.02023 1.27979 1.53178 1.53178 1.53178 1.53178 1.53178 [11,] 1980 0.03562 0.13568 0.29765 0.50595
0.74308 0.99361 1.24540 1.48966 1.48966 1.48966 1.48966 1.48966 [12,] 1981 0.04074 0.14759 0.31438 0.52386
0.75837 1.00306 1.24664 1.48117 1.48117 1.48117 1.48117 1.48117 [13,] 1982 0.03904 0.14384 0.30945 0.51910
0.75513 1.00245 1.24943 1.48784 1.48784 1.48784 1.48784 1.48784 [14,] 1983 0.04161 0.13837 0.28029 0.45148
0.63773 0.82797 1.01428 1.19137 1.19137 1.19137 1.19137 1.19137 [15,] 1984 0.04419 0.15621 0.32800 0.54129
0.77814 1.02377 1.26715 1.50062 1.50062 1.50062 1.50062 1.50062 [16,] 1985 0.04034 0.14934 0.32218 0.54149
0.78880 1.04824 1.30756 1.55806 1.55806 1.55806 1.55806 1.55806 [17,] 1986 0.04170 0.15146 0.32310 0.53894
0.78079 1.03330 1.28477 1.52700 1.52700 1.52700 1.52700 1.52700 [18,] 1987 0.03381 0.13192 0.29352 0.50371
0.74493 1.00130 1.26014 1.51216 1.51216 1.51216 1.51216 1.51216 [19,] 1988 0.03848 0.14476 0.31526 0.53324
0.78035 1.04062 1.30158 1.55426 1.55426 1.55426 1.55426 1.55426 [20,] 1989 0.04368 0.15813 0.33671 0.56094
0.81191 1.07373 1.33432 1.58522 1.58522 1.58522 1.58522 1.58522 [21,] 1990 0.04169 0.15045 0.31974 0.53198
0.76927 1.01663 1.26268 1.49944 1.49944 1.49944 1.49944 1.49944 [22,] 1991 0.03868 0.14203 0.30497 0.51092
0.74253 0.98502 1.22702 1.46051 1.46051 1.46051 1.46051 1.46051 [23,] 1992 0.04000 0.14760 0.31784 0.53353
0.77650 1.03118 1.28560 1.53125 1.53125 1.53125 1.53125 1.53125 [24,] 1993 0.03854 0.14705 0.32294 0.54933
0.80720 1.07976 1.35378 1.61969 1.61969 1.61969 1.61969 1.61969 [25,] 1994 0.03616 0.14699 0.33502 0.58435
0.87442 1.18586 1.50278 1.81325 1.81325 1.81325 1.81325 1.81325 [26,] 1995 0.03846 0.14567 0.31850 0.54016
0.79203 1.05775 1.32451 1.58308 1.58308 1.58308 1.58308 1.58308 [27,] 1996 0.03813 0.14463 0.31651 0.53712
0.78792 1.05260 1.31841 1.57610 1.57610 1.57610 1.57610 1.57610 [28,] 1997 0.04461 0.15178 0.31162 0.50650
0.72015 0.93962 1.15549 1.36139 1.36139 1.36139 1.36139 1.36139 [29,] 1998 0.04000 0.14045 0.29371 0.48336
0.69347 0.91100 1.12624 1.33251 1.33251 1.33251 1.33251 1.33251 [30,] 1999 0.03727 0.14560 0.32421 0.55667
0.82357 1.10733 1.39390 1.67297 1.67297 1.67297 1.67297 1.67297 [31,] 2000 0.03476 0.14508 0.33587 0.59205
0.89273 1.21771 1.55009 1.87700 1.87700 1.87700 1.87700 1.87700 [32,] 2001 0.03317 0.13932 0.32369 0.57195
0.86395 1.18001 1.50364 1.82224 1.82224 1.82224 1.82224 1.82224 [33,] 2002 0.03575 0.14505 0.33023 0.57554
0.86076 1.16684 1.47819 1.78312 1.78312 1.78312 1.78312 1.78312 [34,] 2003 0.03969 0.15386 0.34103 0.58375
0.86169 1.15662 1.45403 1.74331 1.74331 1.74331 1.74331 1.74331 [35,] 2004 0.03763 0.14856 0.33287 0.57393
0.85167 1.14773 1.44733 1.73955 1.73955 1.73955 1.73955 1.73955 [36,] 2005 0.03699 0.14999 0.34138 0.59487
0.88956 1.20577 1.52740 1.84237 1.84237 1.84237 1.84237 1.84237 [37,] 2006 0.03750 0.15243 0.34742 0.60599
0.90680 1.22978 1.55844 1.88040 1.88040 1.88040 1.88040 1.88040 [38,] 2007 0.03767 0.14935 0.33546 0.57935
0.86073 1.16099 1.46508 1.76185 1.76185 1.76185 1.76185 1.76185 [39,] 2008 0.03574 0.14615 0.33432 0.58456
0.87627 1.18997 1.50955 1.82292 1.82292 1.82292 1.82292 1.82292 [40,] 2009 0.03764 0.14960 0.33650 0.58174
0.86491 1.16727 1.47362 1.77273 1.77273 1.77273 1.77273 1.77273 [41,] 2010 0.03872 0.14979 0.33161 0.56717
0.83672 1.12260 1.41077 1.69098 1.69098 1.69098 1.69098 1.69098 [42,] 2011 0.03092 0.14286 0.35106 0.64425
1.00021 1.39474 1.80614 2.21699 2.21699 2.21699 2.21699 2.21699 [43,] 2012 0.03240 0.14476 0.34878 0.63159
0.97114 1.34435 1.73104 2.11527 2.11527 2.11527 2.11527 2.11527 [44,] 2013 0.03240 0.14476 0.34878 0.63159
0.97114 1.34435 1.73104 2.11527 2.11527 2.11527 2.11527 2.11527 [45,] 2014 0.03240 0.14476 0.34878 0.63159
0.97114 1.34435 1.73104 2.11527 2.11527 2.11527 2.11527 2.11527 [46,] 2015 0.03240 0.14476 0.34878 0.63159
0.97114 1.34435 1.73104 2.11527 2.11527 2.11527 2.11527 2.11527 [47,] 2016 0.03240 0.14476 0.34878 0.63159
0.97114 1.34435 1.73104 2.11527 2.11527 2.11527 2.11527 2.11527 [48,] 2017 0.03240 0.14476 0.34878 0.63159
0.97114 1.34435 1.73104 2.11527 2.11527 2.11527 2.11527 2.11527 [49,] 2018 0.03240 0.14476 0.34878 0.63159
0.97114 1.34435 1.73104 2.11527 2.11527 2.11527 2.11527 2.11527 [50,] 2019 0.03240 0.14476 0.34878 0.63159
0.97114 1.34435 1.73104 2.11527 2.11527 2.11527 2.11527 2.11527 [51,] 2020 0.03240 0.14476 0.34878 0.63159
0.97114 1.34435 1.73104 2.11527 2.11527 2.11527 2.11527 2.11527 [52,] 2021 0.03240 0.14476 0.34878 0.63159
0.97114 1.34435 1.73104 2.11527 2.11527 2.11527 2.11527 2.11527

$wt_ind_6 [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13] [1,] 1970 0.0450 0.1710 0.377 0.6420 0.945 1.265 1.587 1.9000 2.196 2.470 2.7210 2.9460 [2,] 1971 0.0450 0.1710 0.377 0.6430 0.946 1.266 1.588 1.9020 2.198 2.472 2.7230 2.9490 [3,] 1972 0.0300 0.1300 0.306 0.5480 0.835 1.148 1.470 1.7890 2.095 2.382 2.6470 2.8870 [4,] 1973 0.0370 0.1470 0.330 0.5680 0.842 1.134 1.430 1.7180 1.991 2.246 2.4780 2.6880 [5,] 1974 0.0380 0.1470 0.326 0.5580 0.825 1.108 1.393 1.6710 1.934 2.178 2.4020 2.6030 [6,] 1975 0.0340 0.1360 0.310 0.5400 0.808 1.095 1.387 1.6740 1.946 2.201 2.4340 2.6450 [7,] 1976 0.0440 0.1600 0.340 0.5670 0.822 1.087 1.351 1.6060 1.845 2.065 2.2660 2.4460 [8,] 1977 0.0320 0.1300 0.294 0.5100 0.760 1.028 1.300 1.5660 1.818 2.054 2.2700 2.4650 [9,] 1978 0.0320 0.1290 0.295 0.5160 0.774 1.050 1.332 1.6080 1.872 2.117 2.3430 2.5470 [10,] 1979 0.0360 0.1380 0.304 0.5180 0.762 1.020 1.280 1.5320 1.770 1.991 2.1930 2.3750 [11,] 1980 0.0360

0.1360 0.298 0.5060 0.743 0.994 1.245 1.4900 1.721 1.934 2.1300 2.3060 [12,] 1981 0.0410 0.1480 0.314 0.5240 0.758 1.003 1.247 1.4810 1.702 1.905 2.0890 2.2550 [13,] 1982 0.0390 0.1440 0.309 0.5190 0.755 1.002 1.249 1.4880 1.712 1.920 2.1080 2.2780 [14,] 1983 0.0420 0.1380 0.280 0.4510 0.638 0.828 1.014 1.1910 1.356 1.507 1.6430 1.7640 [15,] 1984 0.0440 0.1560 0.328 0.5410 0.778 1.024 1.267 1.5010 1.719 1.921 2.1030 2.2670 [16,] 1985 0.0400 0.1490 0.322 0.5410 0.789 1.048 1.308 1.5580 1.794 2.012 2.2110 2.3890 [17,] 1986 0.0420 0.1510 0.323 0.5390 0.781 1.033 1.285 1.5270 1.755 1.965 2.1560 2.3270 [18,] 1987 0.0340 0.1320 0.294 0.5040 0.745 1.001 1.260 1.5120 1.751 1.973 2.1760 2.3590 [19,] 1988 0.0380 0.1450 0.315 0.5330 0.780 1.041 1.302 1.5540 1.793 2.013 2.2150 2.3960 [20,] 1989 0.0440 0.1580 0.337 0.5610 0.812 1.074 1.334 1.5850 1.821 2.038 2.2360 2.4130 [21,] 1990 0.0420 0.1500 0.320 0.5320 0.769 1.017 1.263 1.4990 1.722 1.927 2.1130 2.2800 [22,] 1991 0.0390 0.1420 0.305 0.5110 0.743 0.985 1.227 1.4610 1.680 1.883 2.0680 2.2340 [23,] 1992 0.0400 0.1480 0.318 0.5340 0.776 1.031 1.286 1.5310 1.763 1.976 2.1710 2.3460 [24,] 1993 0.0390 0.1470 0.323 0.5490 0.807 1.080 1.354 1.6200 1.871 2.104 2.3170 2.5080 [25,] 1994 0.0360 0.1470 0.335 0.5840 0.874 1.186 1.503 1.8130 2.109 2.385 2.6380 2.8670 [26,] 1995 0.0380 0.1460 0.318 0.5400 0.792 1.058 1.325 1.5830 1.827 2.053 2.2600 2.4460 [27,] 1996 0.0380 0.1450 0.317 0.5370 0.788 1.053 1.318 1.5760 1.820 2.045 2.2510 2.4360 [28,] 1997 0.0450 0.1520 0.312 0.5060 0.720 0.940 1.155 1.3610 1.553 1.729 1.8890 2.0310 [29,] 1998 0.0400 0.1400 0.294 0.4830 0.693 0.911 1.126 1.3330 1.526 1.703 1.8640 2.0080 [30,] 1999 0.0370 0.1460 0.324 0.5570 0.824 1.107 1.394 1.6730 1.938 2.183 2.4080 2.6110 [31,] 2000 0.0350 0.1450 0.336 0.5920 0.893 1.218 1.550 1.8770 2.189 2.481 2.7500 2.9940 [32,] 2001 0.0330 0.1390 0.324 0.5720 0.864 1.180 1.504 1.8220 2.127 2.412 2.6740 2.9120 [33,] 2002 0.0360 0.1450 0.330 0.5760 0.861 1.167 1.478 1.7830 2.074 2.344 2.5930 2.8170 [34,] 2003 0.0400 0.1540 0.341 0.5840 0.862 1.157 1.454 1.7430 2.017 2.272 2.5040 2.7140 [35,] 2004 0.0380 0.1490 0.333 0.5740 0.852 1.148 1.447 1.7400 2.017 2.275 2.5110 2.7240 [36,] 2005 0.0370 0.1500 0.341 0.5950 0.890 1.206 1.527 1.8420 2.142 2.422 2.6780 2.9110 [37,] 2006 0.0380 0.1520 0.347 0.6060 0.907 1.230 1.558 1.8800 2.187 2.473 2.7350 2.9730 [38,] 2007 0.0380 0.1490 0.335 0.5790 0.861 1.161 1.465 1.7620 2.044 2.306 2.5460 2.7630 [39,] 2008 0.0360 0.1460 0.334 0.5850 0.876 1.190 1.510 1.8230 2.122 2.400 2.6560 2.8880 [40,] 2009 0.0380 0.1500 0.337 0.5820 0.865 1.167 1.474 1.7730 2.057 2.321 2.5630 2.7820 [41,] 2010 0.0390 0.1500 0.332 0.5670 0.837 1.123 1.411 1.6910 1.956 2.203 2.4280 2.6310 [42,] 2011 0.0310 0.1430 0.351 0.6440 1.000 1.395 1.806 2.2170 2.614 2.990 3.3370 3.6550 [43,] 2012 0.0320 0.1450 0.349 0.6320 0.971 1.344 1.731 2.1150 2.485 2.834 3.1560 3.4490 [44,] 2013 0.0320 0.1450 0.349 0.6320 0.971 1.344 1.731 2.1150 2.485 2.834 3.1560 3.4490 [45,] 2014 0.0320 0.1450 0.349 0.6320 0.971 1.344 1.731 2.1150 2.485 2.834 3.1560 3.4490 [46,] 2015 0.0332 0.1456 0.346 0.6214 0.950 1.310 1.682 2.0506 2.405 2.739 3.0466 3.3266 [47,] 2016 0.0332 0.1456 0.346 0.6214 0.950 1.310 1.682 2.0506 2.405 2.739 3.0466 3.3266 [48,] 2017 0.0332 0.1456 0.346 0.6214 0.950 1.310 1.682 2.0506 2.405 2.739 3.0466 3.3266 [49,] 2018 0.0332 0.1456 0.346 0.6214 0.950 1.310 1.682 2.0506 2.405 2.739 3.0466 3.3266 [50,] 2019 0.0332 0.1456 0.346 0.6214 0.950 1.310 1.682 2.0506 2.405 2.739 3.0466 3.3266 [51,] 2020 0.0332 0.1456 0.346 0.6214 0.950 1.310 1.682 2.0506 2.405 2.739 3.0466 3.3266 [52,] 2021 0.0332 0.1456 0.346 0.6214 0.950 1.310 1.682 2.0506 2.405 2.739 3.0466 3.3266

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0.27000 0.35700 0.43400 0.50300 0.57700 0.68900 1.08900 1.97900 [19,] 1988 0.10000 0.12400 0.15900 0.19700 0.23300 0.34200 0.44400 0.51200 0.58800 0.75000 1.01200 1.37200 [20,] 1989 0.05200 0.10300 0.22000 0.24100 0.27800 0.33900 0.46700 0.58500 0.70200 0.77900 0.88000 1.53800 [21,] 1990 0.06400 0.09100 0.15300 0.26400 0.30900 0.37300 0.46100 0.58200 0.69400 0.83500 0.97000 1.59800 [22,] 1991 0.03700 0.10600 0.13200 0.18600 0.27100 0.38100 0.45100 0.54200 0.66700 0.78700 0.90100 1.05300 [23,] 1992 0.06300 0.08300 0.11800 0.17700 0.23900 0.27500 0.40900 0.52400 0.59400 0.70900 0.85100 1.04600 [24,] 1993 0.01100 0.08900 0.12100 0.18100 0.24600 0.32000 0.40800 0.57900 0.71900 0.85300 0.96500 1.17400 [25,] 1994 0.04100 0.08400 0.11200 0.22400 0.27000 0.33600 0.46200 0.64300 0.80800 0.86800 1.05800 1.42100 [26,] 1995 0.07000 0.09800 0.14500 0.19200 0.27000 0.34000 0.42900 0.57700 0.80700 0.96500 1.11500 1.36700 [27,] 1996 0.06100 0.09200 0.15100 0.19100 0.28000 0.35200 0.52400 0.68300 0.94500 1.21600 1.42600 1.47700 [28,] 1997 0.10400 0.10600 0.14600 0.20100 0.26000 0.35500 0.49500 0.68300 0.88400 1.08800 1.46700 1.64700 [29,] 1998 0.08400 0.12800 0.13800 0.17800 0.24800 0.34000 0.54500 0.80600 1.03500 1.24600 1.41200 1.65500 [30,] 1999 0.09000 0.10900 0.13400 0.17400 0.25000 0.33100 0.46500 0.74200 1.02100 1.25800 1.37600 1.77600 [31,] 2000 0.04300 0.06400 0.16300 0.19600 0.25500 0.34600 0.46600 0.75600 0.99900 1.14100 1.22800 1.56300 [32,] 2001 0.06600 0.09800 0.12200 0.17900 0.25800 0.32500 0.46100 0.61400 0.82800 1.07400 1.36000 1.67100 [33,] 2002 0.03100 0.07400 0.13000 0.20000 0.25700 0.32900 0.44500 0.64500 0.88300 1.10200 1.32100 1.64900 [34,] 2003 0.03600 0.08600 0.11700 0.18600 0.24500 0.30700 0.40000 0.56400 0.76800 1.00500 1.20900 1.53700 [35,] 2004 0.03400 0.08000 0.15800 0.19300 0.24700 0.30700 0.38700 0.52800 0.70000 0.89700 1.08700 1.54100 [36,] 2005 0.02900 0.07500 0.11300 0.19600 0.25900 0.31800 0.39900 0.51700 0.64100 0.76700 0.91800 1.29600 [37,] 2006 0.03300 0.07600 0.11600 0.14100 0.26100 0.35000 0.41900 0.51600 0.63100 0.75200 0.92400 1.26300 [38,] 2007 0.08600 0.07400 0.12100 0.17200 0.22600 0.33100 0.43100 0.51000 0.62100 0.75600 0.90300 1.17700 [39,] 2008 0.03600 0.04800 0.06900 0.18600 0.25400 0.31200 0.41600 0.51500 0.60500 0.71900 0.86100 1.14800 [40,] 2009 0.01400 0.04500 0.10900 0.14200 0.25300 0.33000 0.41100 0.53200 0.62500 0.76400 0.88600 1.14400 [41,] 2010 0.01400 0.05200 0.10100 0.17500 0.23700 0.31300 0.41500 0.53900 0.64900 0.78700 0.96400 1.47300 [42,] 2011 0.01900 0.06700 0.10100 0.19000 0.28700 0.35300 0.46600 0.61300 0.77400 0.92300 1.17300 1.51400 [43,] 2012 0.00700 0.01400 0.08200 0.20200 0.26400 0.35300 0.47600 0.55800 0.71100 0.91200 1.14600 1.60000 [44,] 2013 0.05200 0.12500 0.26800 0.26300 0.31000 0.36200 0.43100 0.50700 0.67800 0.72600 0.93600 1.14300 [45,] 2014 0.05200 0.09300 0.21714 0.26648 0.37160 0.46986 0.60255 0.64963 0.74736 0.75274 1.63556 1.72040 [46,] 2015 0.05116 0.11269 0.21576 0.27687 0.35856 0.42755 0.54448 0.63244 0.77686 0.83326 1.36493 1.45751 [47,] 2016 0.05031 0.13238 0.21439 0.28726 0.34551 0.38524 0.48642 0.61525 0.80636 0.91378 1.09431 1.19461 [48,] 2017 0.05621 0.09380 0.44480 0.35273 0.36920 0.43705 0.52466 0.61555 0.65331 0.83729 1.07100 1.11000 [49,] 2018 0.05284 0.10639 0.29211 0.30216 0.36210 0.43072 0.53787 0.62681 0.73568 0.83460 1.26696 1.34167 [50,] 2019 0.12506 0.13132 0.19282 0.41566 0.45122 0.61747 0.74028 0.80397 0.96805 1.03429 1.13205 1.16435 [51,] 2020 0.13184 0.19589 0.21236 0.22292 0.50144 0.67483 0.70558 0.90152 1.06666 1.09025 1.22443 1.16435 [52,] 2021 0.13184 0.19589 0.21236 0.22292 0.50144 0.67483 0.70558 0.90152 1.06666 1.09025 1.22443 1.16435

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1.21602 2.71934 4.81855 [15,] 2020 3.09621 2.04509 4.60382 3.15897 2.06623 2.82035 6.38729 [16,] 2021 12.10529 61.16585 3.18122 3.47861 2.32585 1.17366 5.18878

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4596.139 0.29193 2.34436 [3,] 1972 0.90486 1 0 0.08237 0.13454 0.01108 0.37937 778.2773 4452.741 0.28283 2.34834 [4,] 1973 0.86996 1 0 0.11584 0.13322 0.01543 0.37210 823.1917 4320.463 0.27442 2.36746 [5,] 1974 0.78906 1 0 0.20793 0.12391 0.02576 0.37136 900.6714 4307.015 0.27357 2.32417 [6,] 1975 0.77601 1 0 0.21621 0.13447 0.02907 0.38204 759.2883 4501.361 0.28591 2.19425 [7,] 1976 0.73326 1 0 0.26736 0.13485 0.03605 0.38193 786.9280 4499.298 0.28578 2.21006 [8,] 1977 0.62494 1 0 0.44509 0.11550 0.05141 0.37540 916.6954 4380.539 0.27824 2.33471 [9,] 1978 0.51907 1 0 0.61387 0.13036 0.08003 0.37514 792.1636 4375.760 0.27794 2.39914 [10,] 1979 0.46831 1 0 0.77663 0.12878 0.10001 0.39767 822.8121 4785.683 0.30397 2.23289 [11,] 1980 0.50737 1 0 0.66866 0.12768 0.08538 0.39176 911.8944 4678.185 0.29715 2.35777 [12,] 1981 0.39879 1 0 0.97379 0.13224 0.12877 0.39158 883.8073 4674.973 0.29694 2.35995 [13,] 1982 0.30065 1 0 1.52351 0.14224 0.21671 0.39363 734.4645 4712.259 0.29931 2.19430 [14,] 1983 0.36044 1 0 1.30603 0.13141 0.17162 0.42299 801.8155 5246.340 0.33323 1.98306 [15,] 1984 0.28618 1 0 1.91561 0.13343 0.25561 0.42178 813.2920 5224.352 0.33184 1.97760 [16,] 1985 0.30450 1 0 1.77363 0.13551 0.24035 0.42380 835.9211 5261.158 0.33417 2.03829 [17,] 1986 0.35303 1 0 1.63353 0.12656 0.20674 0.45596 946.3597 5846.059 0.37133 2.06418 [18,] 1987 0.31513 1 0 2.05759 0.12898 0.26539 0.45586 892.2877 5844.405 0.37122 2.28107 [19,] 1988 0.29163 1 0 2.18973 0.14889 0.32604 0.43215 863.5017 5412.902 0.34381 2.52851 [20,] 1989 0.29848 1 0 2.12915 0.15139 0.32234 0.43423 947.5844 5450.728 0.34622 2.46891 [21,] 1990 0.30788 1 0 2.06270 0.15215 0.31385 0.43786 990.1568 5516.806 0.35041 2.37532 [22,] 1991 0.30111 1 0 1.96820 0.18127 0.35678 0.41197 886.6238 5045.761 0.32049 2.46887 [23,] 1992 0.29398 1 0 2.38862 0.17555 0.41932 0.43046 806.4663 5382.208 0.34186 2.15609 [24,] 1993 0.27907 1 0 2.50045 0.16105 0.40269 0.43666 895.5872 5495.001 0.34903 1.95172 [25,] 1994 0.23512 1 0 3.32724 0.14996 0.49897 0.44214 981.8491 5594.697 0.35536 1.63171 [26,] 1995 0.15473 1 0 4.62604 0.13592 0.62875 0.43251 948.8014 5419.538 0.34423 1.24824 [27,] 1996 0.12348 1 0 4.94374 0.12071 0.59675 0.43197 998.1562 5409.707 0.34361 0.89430 [28,] 1997 0.10334 1 0 5.35419 0.11396 0.61018 0.43134 971.3623 5398.287 0.34288 0.67887 [29,] 1998 0.15932 1 0 3.37228 0.11008 0.37123 0.43570 973.3235 5477.510 0.34792 0.68236 [30,] 1999 0.23056 1 0 2.32262 0.10667 0.24775 0.44259 940.7492 5602.874 0.35588 0.80647 [31,] 2000 0.27202 1 0 1.76733 0.10924 0.19307 0.42533 900.8304 5288.852 0.33593 1.05509 [32,] 2001 0.20097 1 0 2.33833 0.10536 0.24638 0.42523 870.0468 5287.116 0.33582 1.19643 [33,] 2002 0.27463 1 0 1.95064 0.11253 0.21951 0.44010 914.6487 5557.588 0.35300 1.25889 [34,] 2003 0.27317 1 0 1.94888 0.11388 0.22193 0.43814 856.1234 5521.988 0.35074 1.32294 [35,] 2004 0.26721 1 0 2.01495 0.11851 0.23879 0.43405 849.8339 5447.478 0.34601 1.28011 [36,] 2005 0.26966 1 0 1.93610 0.12240 0.23698 0.42636 775.3179 5307.625 0.33713 1.17875 [37,] 2006 0.22047 1 0 2.28037 0.12506 0.28519 0.41291 816.6178 5062.965 0.32159 1.03611 [38,] 2007 0.17998 1 0 2.85839 0.12842 0.36708 0.40988 807.1295 5007.747 0.31808 0.80494 [39,] 2008 0.17645 1 0 2.92225 0.12410 0.36265 0.41498 760.0640 5100.618 0.32398 0.60045 [40,] 2009 0.16317 1 0 3.53934 0.12576 0.44510 0.42066 755.5832 5203.916 0.33054 0.41790 [41,] 2010 0.18660 1 0 2.94695 0.11052 0.32569 0.43977 745.7855 5551.687 0.35263 0.32444 [42,] 2011 0.29797 1 0 1.41242 0.11586 0.16365 0.39656 990.0540 4765.486 0.30269 0.39546 [43,] 2012 0.38349 1 0 1.04781 0.12070 0.12647 0.39678 948.0295 4769.421 0.30294 0.44588 [44,] 2013 0.45971 1 0 0.81096 0.12291 0.09968 0.40040 938.5148 4835.408 0.30713 0.49922 [45,] 2014 0.48957 1 0 0.74982 0.12310 0.09230 0.40969 1077.9229 5004.382 0.31786 0.54858 [46,] 2015 0.49220 1 0 0.74182 0.13154 0.09758 0.41452 1000.8761 5092.317 0.32345 0.61202 [47,] 2016 0.50914 1 0 0.67357 0.13741 0.09256 0.40784 910.3866 4970.686 0.31572 0.71691 [48,] 2017 0.52859 1 0 0.66785 0.13075 0.08732 0.42624 909.5196 5305.368 0.33698 0.79159 [49,] 2018 0.53030 1 0 0.67728 0.12958 0.08776 0.43245 1001.0951 5418.507 0.34417 0.97937 [50,] 2019 0.54260 1 0 0.58540 0.14465 0.08468 0.41039 982.5532 5017.185 0.31868 1.41734 [51,] 2020 0.57749 1 0 0.50669 0.14737 0.07467 0.41120 1075.4644 5031.914 0.31961 1.81830 [52,] 2021 0.58093 1 0 0.55889 0.13340 0.07455 0.44009 1053.4311 5557.493 0.35300 1.92005

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4185.500 176 0.350 2256.760 825.2180 7049.11 0.258662 4162.830 177 0.352 2242.200 822.8340 7025.37 0.257862 4140.340 178 0.354 2227.780 820.4550 7001.71 0.257070 4118.030 179 0.356 2213.490 818.0790 6978.12 0.256284 4095.880 180 0.358 2199.330 815.7080 6954.60 0.255506 4073.900 181 0.360 2185.300 813.3410 6931.15 0.254734 4052.090 182 0.362 2171.390 810.9770 6907.77 0.253970 4030.450 183 0.364 2157.600 808.6180 6884.46 0.253212 4008.960 184 0.366 2143.940 806.2630 6861.21 0.252461 3987.640 185 0.368 2130.400 803.9120 6838.04 0.251717 3966.470 186 0.370 2116.980 801.5650 6814.93 0.250979 3945.460 187 0.372 2103.670 799.2210 6791.88 0.250248 3924.600 188 0.374 2090.480 796.8820 6768.91 0.249523 3903.890 189 0.376 2077.400 794.5470 6746.00 0.248804 3883.330 190 0.378 2064.430 792.2160 6723.15 0.248091 3862.910 191 0.380 2051.580 789.8880 6700.36 0.247384 3842.640 192 0.382 2038.830 787.5650 6677.64 0.246684 3822.520 193 0.384 2026.190 785.2450 6654.99 0.245989 3802.530 194 0.386 2013.650 782.9300 6632.39 0.245300 3782.680 195 0.388 2001.220 780.6180 6609.86 0.244616 3762.970 196 0.390 1988.890 778.3100 6587.38 0.243939 3743.400 197 0.392 1976.660 776.0060 6564.97 0.243266 3723.960 198 0.394 1964.530 773.7050 6542.62 0.242600 3704.650 199 0.396 1952.500 771.4080 6520.32 0.241938 3685.460 200 0.398 1940.570 769.1150 6498.09 0.241282 3666.410 201 0.400 1928.730 766.8260 6475.91 0.240632 3647.490 202 0.402 1916.980 764.5400 6453.79 0.239986 3628.680 203 0.404 1905.330 762.2580 6431.72 0.239345 3610.010 204 0.406 1893.770 759.9790 6409.72 0.238710 3591.450 205 0.408 1882.300 757.7040 6387.77 0.238080 3573.010 206 0.410 1870.920 755.4330 6365.87 0.237454 3554.690 207 0.412 1859.620 753.1650 6344.03 0.236833 3536.490 208 0.414 1848.420 750.9000 6322.24 0.236217 3518.400 209 0.416 1837.300 748.6390 6300.51 0.235606 3500.430 210 0.418 1826.260 746.3810 6278.82 0.234999 3482.570 211 0.420 1815.300 744.1270 6257.20 0.234397 3464.820 212 0.422 1804.430 741.8760 6235.62 0.233799 3447.180 213 0.424 1793.640 739.6280 6214.10 0.233206 3429.650 214 0.426 1782.930 737.3840 6192.62 0.232618 3412.220 215 0.428 1772.300 735.1430 6171.20 0.232033 3394.900 216 0.430 1761.750 732.9050 6149.83 0.231453 3377.690 217 0.432 1751.270 730.6700 6128.50 0.230877 3360.570 218 0.434 1740.870 728.4380 6107.23 0.230306 3343.560 219 0.436 1730.540 726.2090 6086.00 0.229738 3326.650 220 0.438 1720.290 723.9830 6064.82

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